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Next Generation Water Prediction Capability (NGWPC)

NGWPC ngenCERF Users Guide

PWS 2.5.1

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-	Refer to cPDM for release date	All	Initial release of document
A	Refer to cPDM for release date	Calibration and Evaluation	Updates reflecting formulation selection release.

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1.0 Introduction

The Next Generation (ngen) tool for Calibration, Evaluation, Reforecast, and Forecast (ngenCERF) GUI contains four (4) workflows: Calibration, Evaluation, Forecast and Verification. Reforecast is also known as hindcasting or verification. The tool can be run using either the ngenCERF GUI or from the Command Line Interface (CLI).

The ngenCERF tool initially used the ngen-cal¹ module for calibration and validation and has since been deprecated in the NGWPC program. It has been refactored into several components, including the calibration manager, evaluation manager, model setup workflow manager and forecast manager. The calibration manager iteratively calls the ngen module with calibration parameters tuned for each iteration and evaluates the output of the ngen run, automatically adjusting the calibration parameters before running ngen again. The program iteratively repeats this process until the stop criteria has been met. Currently the only stop criteria is a user-specified number of iterations per optimization algorithm worker. After the calibration is complete, a final validation run is performed using the best set of parameters determined during the calibration. Generated results are viewable in the Evaluation workflow and available for download to the user's computer.

2.0 Definitions

The following table provides definitions to terms used in the ngenCERF tool and is not exhaustive.

Table 1: Definitions of Terms

Term	Definition
Catchment	A catchment is a hydrology-specific control volume with specific characteristics: a realized flowpath connecting inflow (or headwater) to outflow; and an area bounded by a divide that connects landscape processes to a flow network.
Domain	Geographical area of interest. E.g. CONUS, Alaska, Hawaii, Puerto Rico/US Virgin Islands.
Forcing data	Hydrological and atmospheric forces on a catchment, such as amount of rain, wind velocity, etc.
Formulation	Selection of modules defining the model stack to run in ngen.
Gage	Measures stream flow. For the purpose of calibration, it is also known as the Headwater Basin Gage.
GeoPackage	The hydrofabric file associated with a specific gage. This is often called Subsetted Hydrofabric.
Hydrofabric	A dataset containing a network of connected representations of rivers, lakes, and catchments.

¹ ngen-cal was originally developed by NOAA OWP and updated for NGWPC.

Term	Definition
Data Services	Provides API calls to retrieve forcing, observational, subsetting hydrofabric GeoPackage, and module initial parameters based upon a Gage ID and formulation. This name for this API is still evolving so this document uses the generic term Data Services to refer to data provided by the NGWPC Enterprise Data and FIM Services (THE CALIBRATION MEDFSANAGER) team.
Metrics	Statistics calculated by the calibration manager during calibration.
Objective Function	Metric used as the error function on the output variable.
Observational data	Historical streamflow data observed at a gage.
Optimization Algorithm	Algorithm used to optimize the calibration.
Reference dataset	Output from NWM v3.0.
Subsetting Hydrofabric	The hydrofabric associated with a specific gage. This is also called the GeoPackage.
Tuning Controls	Times, calibratable parameters, and output variable for performing a calibration with ngen-cal.

3.0 Running ngenCERF

There are two (2) ways to run ngenCERF; the Command Line Interface (CLI) and the web-based Graphical User Interface (GUI). To use either of these interfaces, the user must log in to Parallel Works² (PW) and connect to the operational environment shared session. After establishing a connection, the user will be presented with the GUI. It is strongly recommended to click the “Open in Browser” button at the top right to initiate the GUI.

The information required to run a calibration is stored in a database and flat files. This includes subsetting data retrieved from Data Services. The GUI is designed to step the user through the calibration setup process using a series of tabs to gather the required data. After all the required data has been provided or retrieved, the status of the job will be set to *Ready* and can be *Run*. To run calibration using the CLI, a Calibration Definition File is required. This file can be exported from a previously run calibration job and modified for new calibration job(s).

4.0 ngenCERF CLI

The ngenCERF CLI provides a command-line interface to interact with the ngenCERF server, allowing the user to create, update, manage, and run calibration jobs. The CLI is an alternative to the GUI, enabling automation and scripting capabilities. This section lists the commands available through the CLI. See the [ngenCERF CLI Confluence page](#) for the latest command updates.

To run the CLI, log in to Parallel Works, select “Clusters” in the left navigation bar and double click on the OE cluster. In the upper right will be your username followed by the @ symbol and IP address. Click on your username to open a terminal to run the CLI.

4.1 CLI Commands

CLI commands must be prefixed with `ngencertif`, followed by arguments based on the command parameters.

Syntax

`ngencertif <command> <args>`

Legend

`< >` Angle Brackets indicate required information
`[ ]` Square brackets indicate an optional argument

4.1.1 about

Returns the commit hash and commit details for each ngenCERF repository associated with the calibration job.

² Parallel Works is a third-party software, supporting multi-cloud deployment, for provisioning of cloud resources to run and execute HPC type and other workloads for the NGWPC program.

Usage

`ngincerf about <run_id> [--output <location>]`

Arguments

<code>run_id</code>	A single calibration run ID.
<code>--output, -o</code>	(Optional) The path to save the export file. Defaults to current working directory.

Example

`ngincerf about 123`

4.1.2 archive

Archives one or more calibration jobs, removing them from most operations without permanently deleting them.

Usage

`ngincerf archive <run_id>`

Arguments

<code>run_id</code>	A space-separated list of one or more calibration run IDs.
---------------------	--

Example

`ngincerf archive 123`
OR
`ngincerf archive 123 145 201`

4.1.3 cancel

This command will cancel a running calibration job id.

Usage

`ngincerf cancel <run_id>`

Arguments

<code>run_id</code>	Required. Integer number.
---------------------	---------------------------

Example

`ngincerf cancel 123`
OR
`ngincerf archive 123 145 201`

4.1.4 delete

Permanently deletes one or more calibration jobs after prompting for confirmation.

Usage

`ngincerf delete <run_ids>`

Arguments

run_ids A space-separated list of one or more calibration run IDs to delete.

Example

```
ngincerf delete 123
OR
ngincerf delete 123 145 201
```

4.1.5 download

Downloads a ZIP archive for a calibration run.

Usage

```
ngincerf download <run_id> [--output OUTPUT]
```

Arguments

run_ids A space-separated list of one or more calibration run IDs to delete.
--output, -o (Optional) The path to save the export file. Defaults to current working directory.

Example

```
ngincerf download 123
OR
ngincerf delete 123 --output ~/my_results/
```

4.1.6 export

Export an existing calibration from the ngenCERF tool. The exported Calibration Definition File can be used in the **ngincerf import** command or as a template for new calibrations. See the table below for the Calibration Definition File entry descriptions and the Data section for an example Calibration Definition File JSON. The exported filename will be the job ID and username.

Usage

```
ngincerf export <run_id> [--output OUTPUT] [--show]
```

Arguments

run_id Required. Integer number.
--output, -o (Optional) The path to save the export file. Defaults to current working directory.
--show, -s (Optional) Display the job in the console.

Example

```
ngincerf export 123 --output ./job_456.json --show
```

4.1.7 import

Imports a new job definition from a JSON file, with an optional `--run` flag to immediately start the job. If `--run` is specified, it overrides `run_after_import` if specified in the JSON file

Usage

```
ngincerf import input_file [--run true|false]
```

Arguments

<code>input_file</code>	The path to the JSON file containing the Calibration definition file. See the table below and the Data section for an example JSON file.
<code>--run, -r</code>	(Optional) Overrides the run_after_import field in the JSON file. Defaults to true if specified without a value.

Example

```
ngincerf import my_job.json --run false
```

4.1.8 jobs

Lists all calibration jobs, optionally saving the output to a markdown file.

Usage

```
ngincerf jobs [--output OUTPUT]
```

Arguments

<code>--output, -o</code>	(Optional) The path to save the job list. Default is the current working directory.
---------------------------	---

Example

```
ngincerf import my_job.json --run false
```

4.1.9 register

Registers a new username (email) and password for use with ngenCERF. The credentials will be saved in `~/.ngincerf_env`.

Usage

```
ngincerf register [email]
```

Arguments

<code>email</code>	(Optional) The email address to register. If omitted, you will be prompted.
--------------------	---

Example

```
ngincerf register user@example.com
```

4.1.10 run

Run an existing calibration job ID if not already run or not running. This command will only be necessary if the **ngincerf import** argument **run_after_import** was set to false, or the calibration job ID was setup in the GUI and not yet run.

Usage

```
ngincerf run <run_id>
```

Arguments

<code>run_id</code>	Required. Integer number.
---------------------	---------------------------

Example

```
ngincerf register user@example.com
```

4.1.11 show

Displays the details of a calibration job setup.

Usage

```
ngincerf show <run_id>
```

Arguments

<code>run_id</code>	The calibration run ID to display.
---------------------	------------------------------------

Example

```
ngincerf show 123
```

4.1.12 unarchive

Unarchives one or more calibration jobs, restoring them to active status.

Usage

```
ngincerf unarchive <run_ids>
```

Arguments

<code>run_ids</code>	A space-separated list of one or more calibration run IDs.
----------------------	--

Example

```
ngincerf unarchive 1 2 3
```

4.1.13 update

Updates an existing calibration job that has not been run with new parameters from a JSON file.

Usage

```
ngincerf update <run_id> <input_file>
```

Arguments

<code>run_ids</code>	Calibration run ID to update
<code>input_file</code>	The path to the JSON file containing the updated job parameters

Example

```
ngincerf update 456 updated_job.json
```

4.1.14 upload-forcing

Uploads forcing data files for a calibration run.

Usage:

```
ngincerf upload-forcing <run_id> <forcing_dir>
```

Arguments:

<code>run_id</code>	The calibration run ID.
<code>forcing_dir</code>	The directory containing forcing data files to upload.

Example:

```
ngincerf upload-forcing 456 ./forcing_data
```

4.1.15 upload-geopkg

Uploads a GeoPackage file for a calibration run.

Usage:

```
ngincerf upload-geopkg <run_id> <geopkg_file>
```

Arguments:

<code>run_id</code>	The calibration run ID.
<code>geopkg_file</code>	The path to the GeoPackage file to upload.

Example:

```
ngincerf upload-geopkg 456 ./data.gpkg
```

4.1.16 upload-obs

Uploads observation data files for a calibration run.

Usage:

```
ngincerf upload-obs <run_id> <obs_dir>
```

Arguments:

<code>run_id</code>	The calibration run ID.
<code>obs_dir</code>	The directory containing observation data files to upload.

Example:

```
ngincerf upload-obs 456 ./obs_data
```

4.2 Output Files

Several subcommands save the output to disk. By default, files are saved to **the current working directory** using a default name that varies by command.

You can override the default location and/or filename using the optional **--output** or **--export** argument (depending on the command). The behavior is as follows:

- **Directory only:** If you specify a directory (e.g., **--output ~/foo/**), the default filename is used within that directory.
- **Filename only:** If you specify a filename without a directory (e.g., **--output new-file.txt**), the file is saved to **current working directory** with the given name.
- **Full path:** If you specify a full path (e.g., **--output ~/foo/new-file.txt**), the file is saved exactly at that location.

Examples:

- **--output ~/** → saves to **~/my-default.txt**
- **--output ~/foo/** → saves to **~/foo/my-default.txt**
- **--output new-file.txt** → saves to **<current working directory>/new-file.txt**
- **--output ~/foo/new-file.txt** → saves to **~/foo/new-file.txt**

4.3 Importing and Exporting

4.3.1 Calibration Definition File

The Calibration Definition File is a JSON file containing the necessary entries to completely setup and run a calibration. It can be generated from an existing calibration job and used directly as input for a new calibration job or used as a template to create a new calibration setup.

The metadata section of this file is ignored for the **ngencerf import** command. The user may add information in this section needed for reference.

See **Section 6.5** for an example JSON File.

Table 2: Calibration Definition File

Entry	Description
source_calibration_run_id	Metadata: ID of the job that was exported.
time_range	Metadata: Intersection of time range from the forcing data and observation data, constraining calibration and validation.
errors	Metadata: Shows any outstanding errors in the job

Entry	Description
	that would prevent it from running.
run_after_import	Flag indicating whether to submit the job immediately after import and any necessary uploads, provided there are not any errors. Obtained from the CLI command.
gage_id	The gage_id to calibrate.
agency	The owner of the gage.
The paths specified for forcing, observational and GeoPackage data do not refer to the user's local machine, but are paths on the server machine. They are obtained from the job which was exported and normally should not be changed, except to clear them. They are only used if the source is User Upload.	
forcing_source	The source of the forcing data. Currently, only Upload is supported, indicating that the file needs to be uploaded by the user.
forcing_user_uploaded_dir_path	The location of the forcing data uploaded by the user to the exported job.
observational_source	The source of the observational data, e.g., Data Services or User Upload.
observational_user_uploaded_dir_path	The location of the observational data uploaded by the user to the exported job.
geopackage_source	The source of the GeoPackage data. E.g., Data Services or User Upload
geopackage_user_uploaded_file_path	The location of the GeoPackage file uploaded by the user to the exported job.
modules	The list of modules used for this calibration.
formulation_name	User-supplied formulation name. This has no effect on the job.
use_sloth	Flag indicating the SLoTH model will be added to the formulation and sloth_parameters must be supplied. Note: <i>The current calibration manager design does not allow for specification of SLoTH variables.</i>
sloth_parameters	Required if use_sloth is true. The SLoTH parameter definitions. All SLoTH fields are required for each SLoTH variable. Note: <i>The current calibration</i>

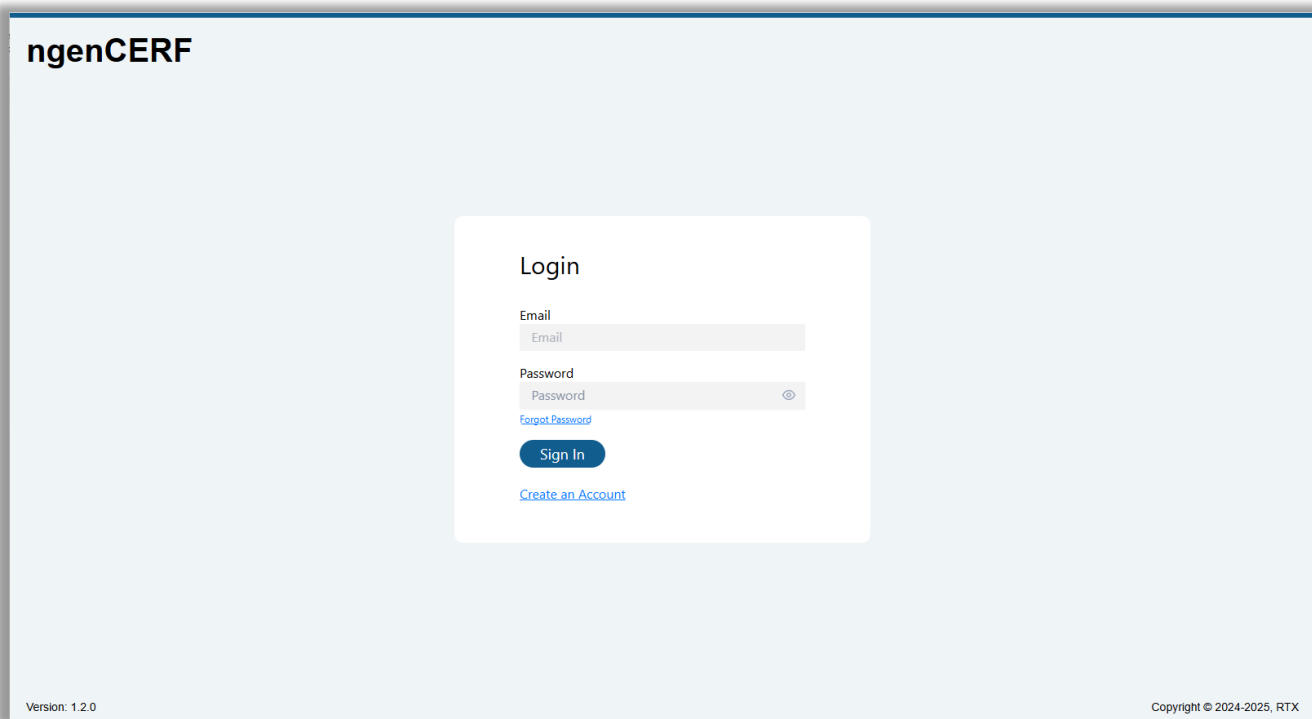
Entry	Description
	<i>manager design does not allow for specification of SLoTH variables.</i>
automatic_validation	If true, a validation run is executed automatically after the calibration run. Note: <i>Due to the current calibration manager design, this will always true.</i>
calibration_times	The time ranges to use for calibration.
validation_times	The time ranges to use for validation (Must be outside the calibration times).
stream_flow_threshold	Required if a categorical objective function is specified otherwise, if empty, the calibration manager will not calculate categorical metric.
peak_flow_threshold	Required if an event-based objective function is specified otherwise, if empty, the calibration manager will not calculate event-based metric.
parameters	Module parameters to use for calibration tuning including name, min, max, and module. All fields are required.
objective_function	Metric to use for the objective function.
optimization	Optimization algorithm (DDS, GWO, PSO).
optimization_inputs	Inputs for the selected optimization.
save_plot_iteration_frequency	How often the calibration manager will generate plots during the calibration.
save_output_iteration	If true, output for each iteration is saved in separate files (not supported by UI).
stop_criteria	Number of worker iterations to run.

5.0 ngenCERF GUI

The GUI is a series of pages with tabbed work areas for setting up a calibration, validation, forecast, and verification (aka reforecast or hindcast). After logging in, along the top of the display, are the Workflows of the tool, a help button and a user profile button. The help provides contextual help for each page or tabbed work area. A summary area on the right side of the workflow page provides progress of the workflow setup along with details of the job setup.

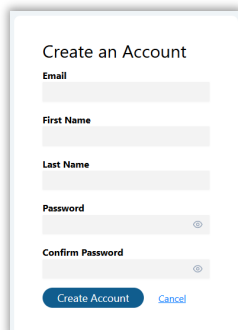
5.1 Login Page

Enter your ngenCERF username and password or request a ngenCERF account.



The screenshot shows the ngenCERF login interface. At the top left, the text "ngenCERF" is displayed. In the center, there is a white box titled "Login". Inside this box, there are two input fields: "Email" and "Password". Below the "Password" field is a link that says "Forgot Password". At the bottom of the login box is a blue "Sign In" button and a link that says "Create an Account". In the bottom left corner of the page, it says "Version: 1.2.0". In the bottom right corner, it says "Copyright © 2024-2025, RTX".

To create an account, provide a username, password, and email.

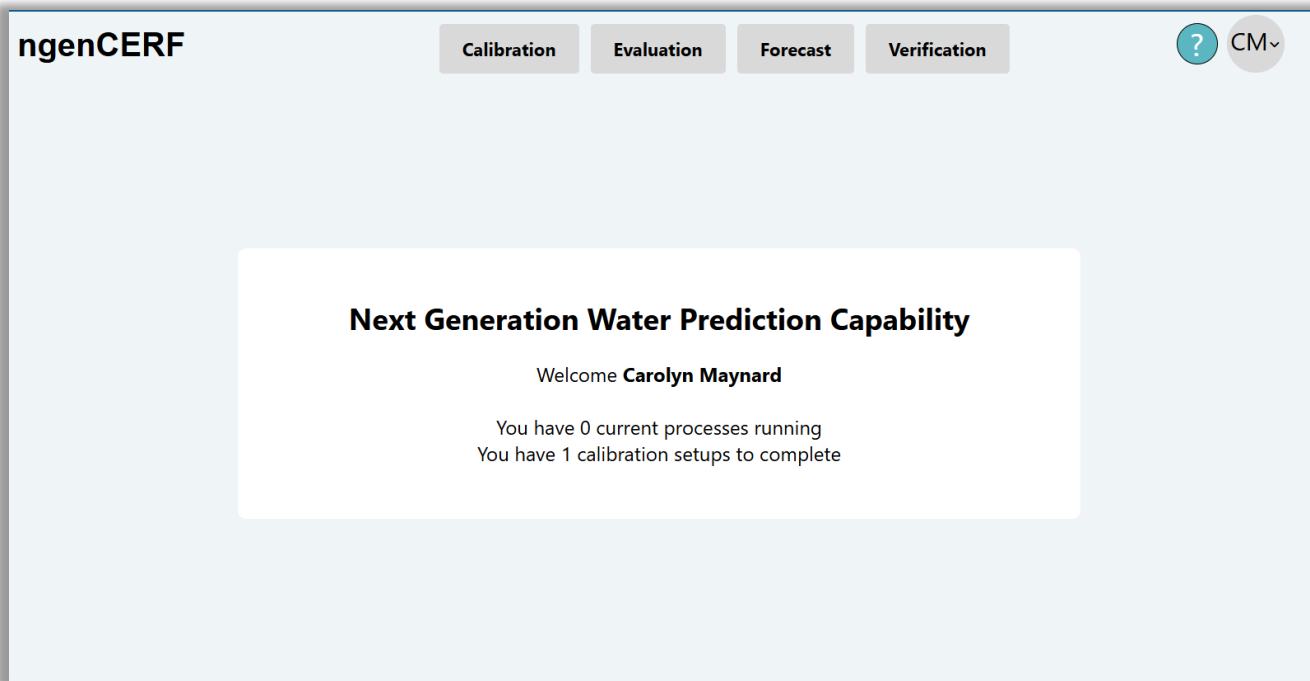


The screenshot shows the "Create an Account" form. It has five input fields: "Email", "First Name", "Last Name", "Password", and "Confirm Password". Below the "Password" and "Confirm Password" fields are eye icons to toggle visibility. At the bottom of the form are two buttons: a blue "Create Account" button and a "Cancel" link.

To Logout, click your initials at the top of the page and select Logout.

5.2 Landing Page

After a successful login, this page will display any running or incomplete calibrations. The header contains the different workflows for the ngenCERF tool.



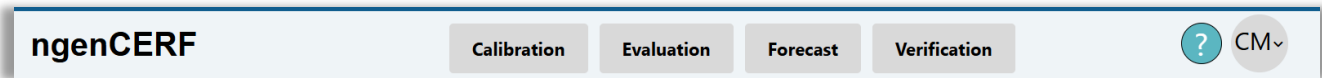
Header Workflow Buttons:

Calibration	Select to move to the Calibration workflow.
Evaluation	Select to move to the Evaluation workflow. This is only available when at least one calibration job has successfully completed.
Forecast	Select to move to the Forecast workflow. This is only available when at least one calibration job has successfully completed.
Verification	Select to move to the Verification workflow. This is a beta version of the capability that will be available in the next release.

5.3 ngenCERF Interface Basics

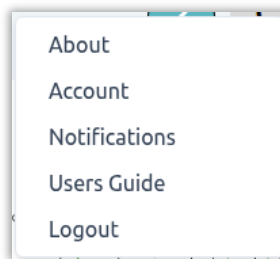
ngenCERF is designed with four (4) workflows, Calibration, Evaluation, Forecast, and Verification. After a workflow button is clicked the work area is displayed with a series of tabs based on the workflow. Contextual help is available by clicking the question mark in the header area. Right-click context menus are used for actions associated with a calibration job in a table.

5.3.1 ngenCERF Header Area



The header area contains the tool name, the Workflow buttons, the help button and the avatar button. The help button provides contextual help based on the Workflow and tab. When the avatar button is clicked a submenu is displayed with entries for About, Account, Notifications and Logout.

Avatar button submenu:



Options:

About

Displays table with the commit hash for all repositories associated with ngenCERF release.

Account

Opens User Profile page to update user details, including password.

Notifications

Provides a table of all notification messages generated during the users ngenCERF session.

Users Guide

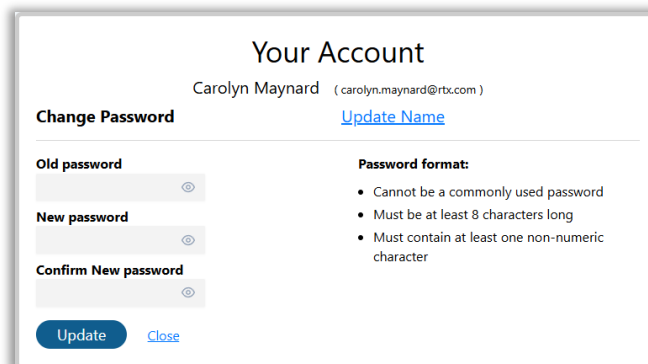
Opens the Users Guide in a separate browser tab for viewing and downloading.

Logout

Logs the user out of ngenCERF.

5.3.2 User Profile Page

Use this page to update your username and password.

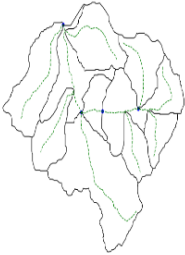


5.3.3 Calibration Progress Area

The Calibration workflow only has an area on the right side of the page showing progress made towards completing a calibration setup. It will appear after the user either clicks “New” or selects an existing calibration job to open from the Calibration Runs tab.

Progress

- ✓ [Headwater Basin Gage](#)
- ✓ [Forcing](#)
- ✓ [Observational](#)
- ✓ [Geopackage](#)
- ✓ [Formulation](#)
- ✓ [Start and End Times](#)
- ✓ [Output Variable to Calibrate](#)
- ✓ [Tuning Parameters](#)
- ✓ [Optimization Algorithm](#)
- ✓ [Objective Function](#)
- ✓ [Calibration Stop Criteria](#)
- ✓ [Metrics / Plot Interval](#)



Setup - Calibration Job ID 260

Gage: 12114500
Catchments: 10
Forcing Data: AORC
Observational Data: Data Services

CEDAR RIVER BELOW BEAR CREEK NEAR
CEDAR FALLS, WA

Formulation Name: snow17-sacma-noah-troute-2lter
Modules: Sac-SMA, Noah-OWP-Modular, Snow-17, T-Route

Calibration Run		Validation Run	
Sim Start:	2013-01-01 00:00	Sim Start:	2014-01-01 00:00
Sim End:	2015-04-30 23:00	Sim End:	2016-05-18 23:00
Calib Start:	2014-01-01 00:00	Val Start:	2015-05-01 00:00
Calib End:	2015-04-30 23:00	Val End:	2016-05-18 23:00

Output Var: Streamflow
Optimization Algorithm: DDS
Calibration Stop Criteria: 2

Tuning Parameters: 30
Objective Function: KGE
Plot Generation Frequency: 1

Results Pathname: /ngencertf/data/ngen-cal-work/run_calib/260_cmaynard
NGen Global Logging: ENABLED

As the user progresses through the tabs and saves information, checkmarks are placed next to the required item. After all the items have checkmarks, the calibration job will be in the *Ready* state and can be run.

Note: The calibration items in the Progress area are hyperlinks that, when clicked, will take the user to the tab where this item can be viewed and, if the job has not been run, updated, and saved.

5.4 Calibration Workflow

The calibration workflow allows the user to set up and run a calibration automatically followed by a validation run on the best parameter set determined during the calibration process. Metrics are calculated during each iteration of the calibration and available for examination by the user as the calibration job is running as well as after completion.

During the calibration run, ngen is iteratively executed with a new set of parameter values, re-calibrated based on the previous run to produce a simulated result in the selected output variable to match as closely as possible to the observed value of the output variable. Currently the only output variable available is streamflow, since that is the observed data currently available to the calibration manager.

An automatic validation run executes after a calibration run is complete by running ngen once during a time frame different from the calibration period using the best set of parameters determined during the calibration run.

The Calibration Workflow Interface provides the user with the tools necessary to select the headwater basin to calibrate based on a gage, the formulation of modules (ie. The Model Stack) to use during the calibration, the calibration time period, the validation time period, the output variable to calibrate against (at this time only flow is available), the parameters used for calibration tuning, the optimization algorithm, the objective function, ancillary metrics, calibration stop criteria, plot generation frequency and Error Warning and Trapping System (EWTs) settings.

The forcing data, observational data, and subsetted hydrofabric GeoPackage can be uploaded by the user or obtained from the NGWPC Enterprise Data and FIM Services (THE CALIBRATION MEDFSANAGER) server, referenced as Data Services in this document, based on the gage selected by the user. The set of possible calibratable parameters will be obtained from Data Services based on modules selected for the formulation and the user will be able to select the parameters used for calibration tuning from this list or upload a set of parameters from a file.

The Progress area to the right of the GUI provides the user with a visual cue of the calibration setup progress and running summary of the calibration details.

After all necessary setup information has been provided, the status is set to *Ready* and the user can start the calibration and validation run.

The Calibration Workflow has a series of tabs for setting up a calibration:

- **Calibration Runs** – Lists users calibration jobs
- **Headwater Basin Gage** – Select the headwater basin gage, forcing, and observational data
- **Formulation** – Select the model stack and enter provide a name for the formulation
- **Tuning Controls** – Times, Output Variable to Calibrate, and Calibration Tuning Parameters
- **Optimization / Metrics** – Optimization Algorithm, Objective Function, Additional Metrics, Stop Criteria, and Plot Generation Frequency
- **Run / Status** – Review calibration setup, run, monitor, and view result

Table 3: Available Calibration Setup Items

Item	Description	Tab Location
Domain	Select the domain of interest to narrow the list of gages.	Headwater Basin Gage
Gage	Select the Headwater Basin Gage ID.	Headwater Basin Gage
Forcing Data	Choose to upload the data from the user's computer or use AORC data.	Headwater Basin Gage
Observational Data	Choose to upload the data from the user's computer or retrieve it from the gage agency using Data Services.	Headwater Basin Gage

Item	Description	Tab Location
GeoPackage	Choose to upload the data from the user's computer or retrieved it from Data Services.	Headwater Basin Gage
Formulation Name	User's name for this formulation.	Formulation
Formulation	Selection of modules for the calibration run.	Formulation
Calibration Time Controls	Set of four (4) times defining the length of the simulation and when to start and end the calibration activities within the simulation time.	Tuning Controls
Validation Time Controls	Set of four (4) times defining the length of the simulation and when to start and end the validation activities within the simulation time.	Tuning Controls
Output Variable To Calibrate ³	Currently ngen calibration only supports Streamflow.	Tuning Controls
Calibration Tuning Parameters ³	The set of calibratable parameters from the modules in the formulation selected by the user the calibration manager modifies to fine tune the calibration.	Tuning Controls
Optimization Algorithm ³	The optimization algorithm to be used for the calibration.	Optimization / Metrics
Calibration Stop Criteria ³	The number of ngen iterations for each optimization worker.	Optimization / Metrics
Objective Function ³	A metric to use as the objective function.	Optimization / Metrics
Additional Metrics ³	Optional. Calculate optional Categorical or Event Based Metrics.	Optimization / Metrics
Plot Generation Frequency ³	Determines how often to generate plots during the calibration. Each plot is cumulative. One (1) is every iteration, five (5) is every five iterations, etc.	Optimization Metrics
Global Logging	Enable or Disable the ngen Error Warning and Trapping System Logging	Status / Run
Log Levels	Set the Error Warning and Trapping System log level for ngen, forcing, and each individual module in the formulation.	Status / Run

³ Not available in formulations with LSTM

After the user either selects a previously run calibration or clicks the New button, navigation through the workflow is accomplished using the **Next** and **Prev** buttons in a tab or by clicking a tab name.

Below is the typical flow a user would follow to set up a calibration run. The trapezoids represent the different tabs available for manual entry of calibration data in the Calibration GUI. Below each trapezoid are the items set in the tab.

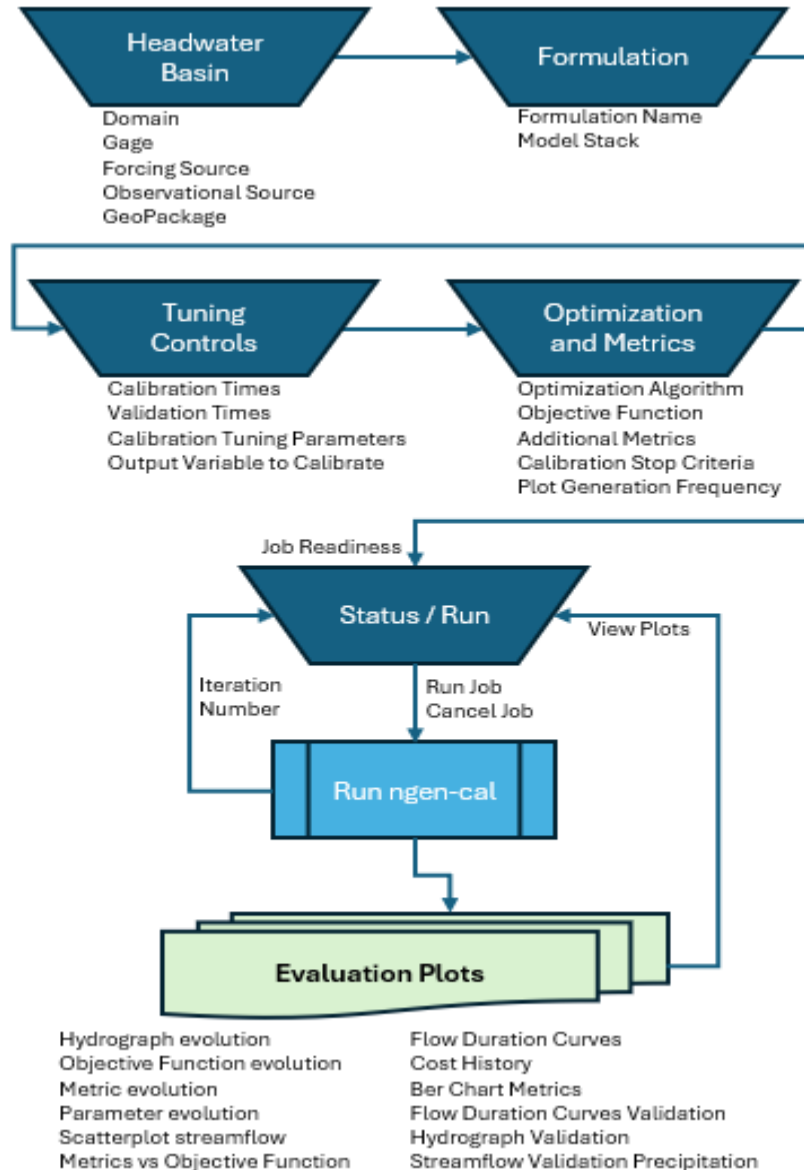


Figure 1: GUI Calibration Setup Workflow

5.4.1 Calibration Runs Tab

The Calibration Runs tab is the landing page when the **Calibration** button in the header area is clicked. This tab allows the user to create a new calibration from scratch or clone a previous calibration to use as a baseline for a new calibration. The user may select other available actions by right-clicking on a calibration job.

All calibration jobs associated with the user are provided regardless of status. A table is displayed showing key information about each calibration job. The tab provides the ability to filter the jobs in the table based on several criteria: Headwater Basin Gage, Module(s), Status(es), and Archived. Each column in the table is sortable by clicking the “sort” icon near the column header.

Calibration Jobs

New

Double click on a row to open, or right click for more options. Click "New" button for a fresh setup.

Headwater Basin Gage

All

Status

▼

Modules

▼

☐ Include Archived

Apply Filters

[Clear Filters](#)
[Refresh List](#)

Job ID	Headwater Basin Gage	Formulation Name	Stop Criteria (Iterations)	Job Genesis	Creation Date	Last Updated	Submit Date	Calibration Period	Status
583	01123000	topmodel-smp-sft-noah-2iter	2	clone	2025-11-12 22:48	2025-11-13 18:38		2015-10-01 00:00 to 2017-09-30 23:00	Ready
582	01123000	topmodel-smp-sft-noah-2iter	2	clone	2025-11-12 19:53	2025-11-12 19:53	2025-11-12 19:53	2015-10-01 00:00 to 2017-09-30 23:00	Done
576	01123000	topmodel-smp-sft-noah-2iter	2	clone	2025-11-11 22:16	2025-11-11 22:16	2025-11-11 22:16	2015-10-01 00:00 to 2017-09-30 23:00	Done
575	01123000	topmodel-smp-sft-noah-2iter	2	clone	2025-11-11 21:54	2025-11-11 21:55	2025-11-11 21:55	2015-10-01 00:00 to 2017-09-30 23:00	Done
574	01123000	topmodel-smp-sft-noah-2iter	2	clone	2025-11-11 20:41	2025-11-11 20:42	2025-11-11 20:42	2015-10-01 00:00 to 2017-09-30 23:00	Done
573	01123000	topmodel-smp-sft-noah-2iter	2	clone	2025-11-10 19:58	2025-11-10 20:01	2025-11-10 20:01	2015-10-01 00:00 to 2017-09-30 23:00	Done

The user can double click on a job to open it, select one or more calibration jobs and right click for available actions. To de-select a job, click on it again.

User Actions:

New

Select this button to create a calibration setup from scratch.

Filters

Filter by Headwater Basin Gage, multi-select status, multi-select modules, and include or exclude archived jobs. The user must click **Apply Filters** to filter the table. The Multi-select filters will show any job that matches one of the selected items. i.e. *It is an “or” filter not an “and” filter.*

Double Click

Opens calibration job and takes user to the Status / Run tab.

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Right Click

Open, Clone, Download, Delete, Archive, or Unarchive the selected calibration job in the Calibration Runs table. The Download option appears for Done jobs only. If the job is already archived, the Unarchive option will be available.

Open

Opens calibration job and takes the user to the Status/Run tab.

Clone

This will make a copy of the calibration job set up as a new calibration job for editing.

Download

Download all data associated with a finished calibration job.

Export

Export the Calibration Definition File. (Used by the CLI.)

Delete

Permanently deletes the selected calibration job(s) from the system.

Archive/Unarchive

Archive or Unarchive the selected calibration job(s). Archived jobs Will no longer appear in any Calibration Runs table unless the Archived filter is active.

Table 4: Calibration Job Status Descriptions

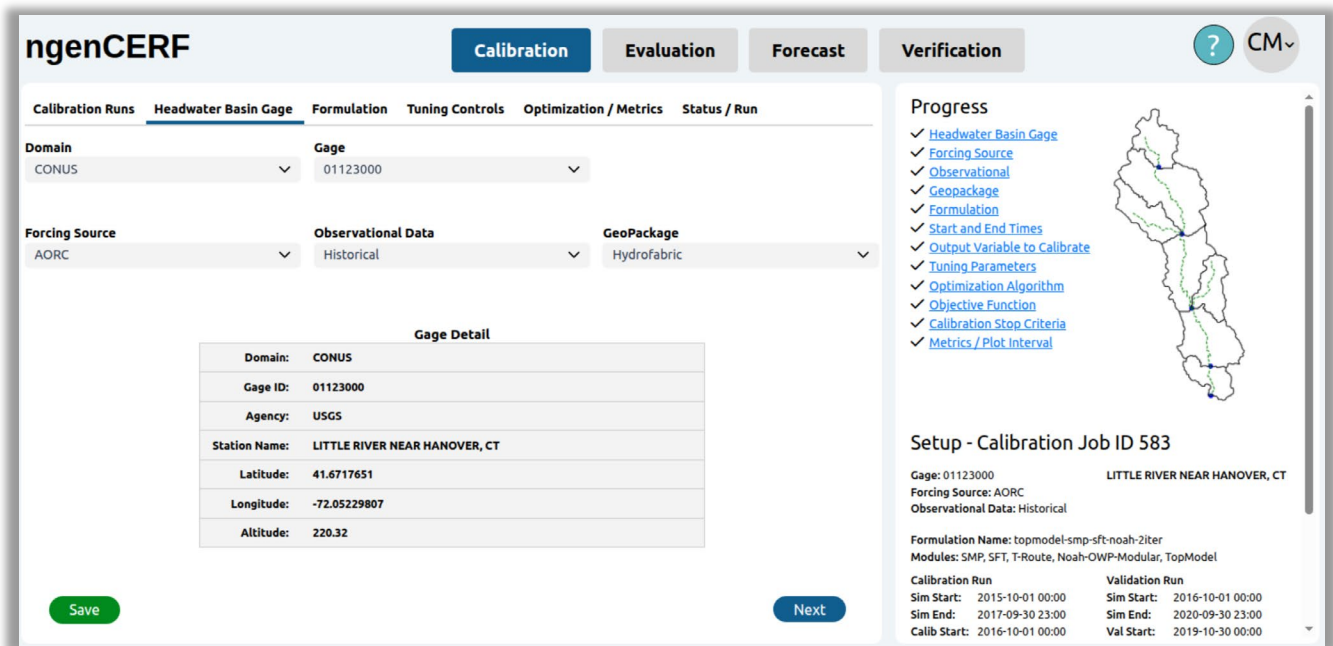
Status	Description
Saved	Some information has been saved, but the setup is not complete. Check the Progress checkmarks for missing items. Click on the item to go to the appropriate tab.
Ready	All the necessary information has been provided, and the job is ready to Run. Click the Run button to start the run.
Validating and Preparing Job Data	The backend is subsetting the data to the selected time range for the calibration/validation runs, running processes that validate/create the ngen realization file and creating the calibration manager input config file to submit the job for execution.
Submitted	Calibration has been submitted and waiting for resources from the cluster to begin execution.
Calibration Running	The calibration has begun. The iteration number reflects the last completed iteration run of ngen. Iteration 0 is the control run executed by the calibration manager for metric calculations. Logs are viewable once the calibration is in a running state. Calibration plots are viewable once iteration 1 is complete. Validation plots are viewable once Validation Best is done.
Calibration Done, Validation Control Running	The calibration has completed. The Calibration Stop Criteria has been reached.. The calibration manager is currently running the control run for validation metric calculations.
Calibration Done, Validation Best Running	Both the calibration run, and validation control run have completed successfully. The validation best is running ngen with the best set of parameters determined during the calibration phase.
Done	Calibration and Validation runs of ngen have successfully completed. Results are viewable in both the Calibration->Status/Run tab and the Evaluation workflow.
Calibration Failed	A failure occurred causing the Calibration Job to crash. Logs and Plots are viewable in the Calibration->Status/Run tab. If running from within the GUI, the ngen.log file will be automatically displayed in the UI. The user can also select to view the logs from the stdout/stdout runs of ngen and the calibration manager, if they exist. Logs and Plots are viewable in the Calibration->Status/Run tab.
Calibration Done, Validation Control Failed	A failure occurred causing the Validation Control Job to crash. Logs and Plots are viewable in the Calibration->Status/Run tab. If running from within the GUI, the ngen.log file will be automatically displayed in the UI. The user can also select to view the logs from the stdout/stdout runs of ngen and the calibration manager, if they exist.
Calibration Done, Validation Best Failed	A failure occurred causing the Validation Best Job to crash. Logs and Plots are viewable in the Calibration->Status/Run tab. If running from within the GUI, the ngen.log file will be automatically

	displayed in the UI. The user can also select to view the logs from the stdout/stdout runs of ngen and the calibration manager, if they exist.
Server Error	The server failed interrupting the Calibration Job either during calibration, or one of the validation runs.
Cancelled	Calibration Job was cancelled by the user.

Note: A Calibration Job in this table refers to both the calibration run and the validation run.

5.4.2 Headwater Basin Gage Tab

The Headwater Basin Gage tab provides the ability to select the headwater basin gage, the historical forcing data, the historical observational streamflow data, and the Subsetted Hydrofabric GeoPackage. The historical and GeoPackage data will be retrieved from Data Services based on the gage and source selections on this tab, but the user is also provided with the capability to upload their own data. The gages provided are those identified by OWP as calibratable headwater basin gages.



User Actions:

Domain

Select a domain to filter the list of gages in the Gage field. Options are CONUS, Alaska, Hawaii, and Puerto Rico/US Virgin Islands.

Gage

Select or enter the gage ID for the Headwater Basin of interest.

Forcing

AORC (default) or Upload⁴. Currently, only AORC data is available from Data Services. If upload is selected, user must provide a CSV file of forcing data. See Table 6: Calibration

⁴ A gage must be selected first to upload Forcing, Observational or GeoPackage data

	Forcing Data Items (in section 6) for a sample of the expected forcings that must be in the CSV file.
Observational	Historical (default) or Upload ⁴ . Data Services use the selected gage's agency as the source for the historical observational data. If upload is selected, the user provides a CSV file of observational data for the entire simulation time range (set on the Tuning Controls tab). See Section 6 for sample Observational Data CSV file.
GeoPackage	Hydrofabric or Upload ⁴ . Data Services will provide the subsetted hydrofabric GeoPackage associated with the selected headwater basin gage.
Save	Click this button to save the data entered on the tab. This can be done periodically or after making all the selections.
Revert	Click this button to revert selections for this tabbed work area to the saved data. It appears if a saved selection changes.
Next	Move to the Formulation Tab. If there is any unsaved data, the user will first be prompted to save.

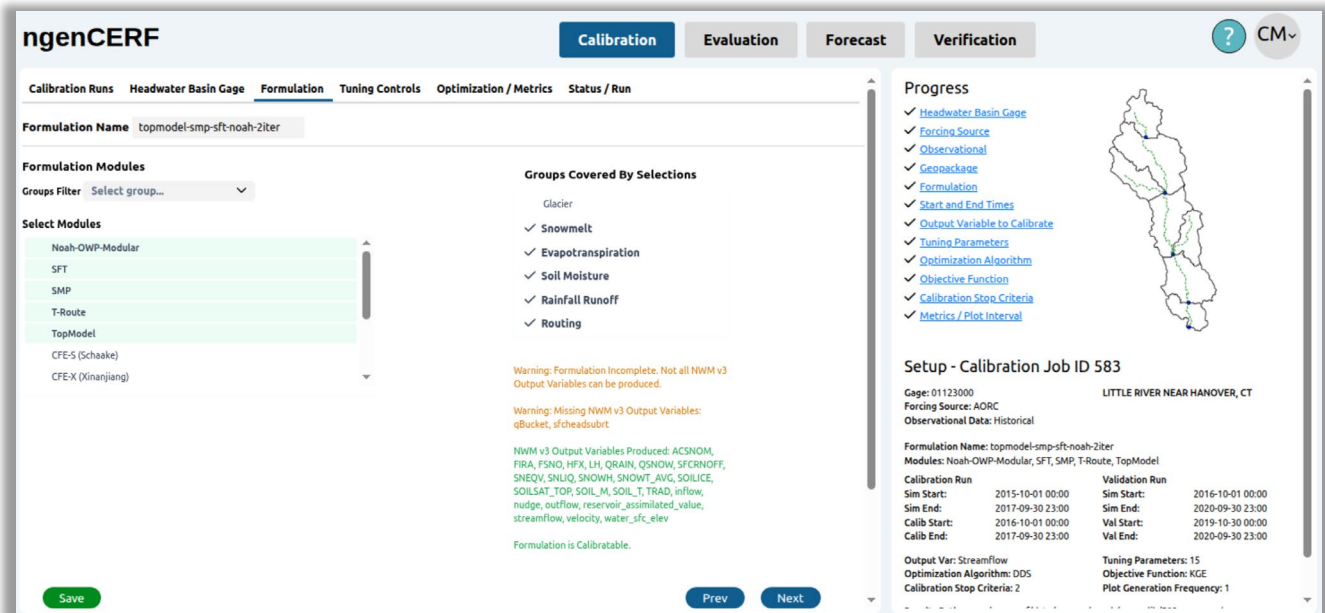
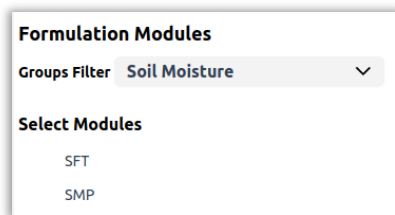
When the Save button is clicked Data Services is called to retrieve the subsetted GeoPackage, Forcing, and Observational data. The appropriate checkmarks are set in the Progress area when the data on this tab has been received from Data Services and is successfully saved.

If you cannot find a gage, verify you have selected the correct domain.

See the [NGWPC Headwater Basin Gage Confluence Page](#) for the list of over 1600 Headwater Basin Gages which includes the gage domain, NWS ID, NWM Reach ID, Agency, Station Name.

5.4.3 Formulation Tab

The Formulation tab provides the ability to select the stack of hydrological modules to use for the calibration and validation runs. Modules are classified by what they model: Glacier, Snowmelt, Evapotranspiration, Rainfall Runoff, Soil Moisture, Channel Routing. When a user selects a module, its associated group(s) are highlighted. This provides the user with an indication of the hydrologic processes covered by the selections. When the user saves the formulation, some basic checking is done on the completeness of the formulation and a warning issued when necessary. The user must have a routing module and at least one module to save a formulation. The completeness of the formulation does not prohibit the calibration from running.

User Actions:

Formulation Name

User's name for the formulation.

Group

Filter the module list by the group classification.

Select Modules

User selects the modules for the formulation.

Save

Click this button to save the data entered on the tab. This can be done periodically or after making all the selections.

Revert

Click this button to revert selections for this tabbed work area to the saved data. It appears if a saved selection changes.

Prev

Move to the Headwater Basin Tab. If there is any unsaved data, the user will first be prompted to save.

Next

Move to the Tuning Controls Tab. If there is any unsaved data, the user will first be prompted to save.

A formulation is considered “Calibratable” if it has an evapotranspiration module (except for LSTM), a rainfall runoff module and a routing module.

A formulation is considered “Complete” if it produces the required NWM v3 Output Variables.

Formulation Rules:

See also [Formulation Rules Confluence Page](#)

- Only one module per process (except for SFT and SMP, which are always run together currently)
- Processes are Evapotranspiration, Snowmelt, Rainfall Runoff, Auxiliary Soil (aka Soil Moisture), and Routing

Evapotranspiration Processes (Required, except for LSTM)

- noah-owp-modular:
 - If noah-owp-modular is selected for evapotranspiration, it is also selected as the snow model if another single purpose snow model has not been selected.
 - If noah-owp-modular has been selected as the snow model, and another single purpose evapotranspiration module has been selected, warn user that there are two evapotranspiration modules selected and noah-owp-modular will not be used for evapotranspiration.
- potential-evapotranspiration (PET): evapotranspiration
 - Not currently integrated in ngenCERF

Snowmelt Processes (Optional)

- Utah Energy Balance (UEB):
 - If selected, will override snowmelt component of noah-owp-modular
 - Cannot be selected in conjunction with another single purpose snow model
- SNOW-17
- noah-owp-modular:
 - If noah-owp-modular is selected for snowmelt, it is also selected as the evapotranspiration module if another single purpose evapotranspiration module has not been selected.
 - If noah-owp-modular has been selected as the evapotranspiration module, and another single purpose snow-melt module has been selected, user is warned that there are two snow-melt modules selected and noah-owp-modular will not be used for snowmelt.

Rainfall Runoff Processes (Required)

- CFE-S, CFE-X
- SAC-SMA
- Topmodel
- LASAM
- LSTM

- If LSTM is selected as the rainfall runoff model the snow-melt, evapotranspiration, and auxiliary soil processes will be disabled.

Auxiliary Soil Processes (Optional)

- SoilFreezeThaw (SFT)
- SoilMoistureProfiles(SMP)
 - SFT and SMP currently run together in the calibration manager. They require either CFE-(X, S) or LASAM. They share parameters with CFE and LASAM, so calibration of those parameters is done jointly.

Routing (Required)

- T-Route

5.4.4 Tuning Controls Tab

The Tuning Controls tab provides the ability to select data that will control how calibration(s) are run. This includes parameters that will be used for calibration tuning, the module output variable to calibrate, the range of time for calibration, and an automatic validation run on the best set of parameters determined during calibration.

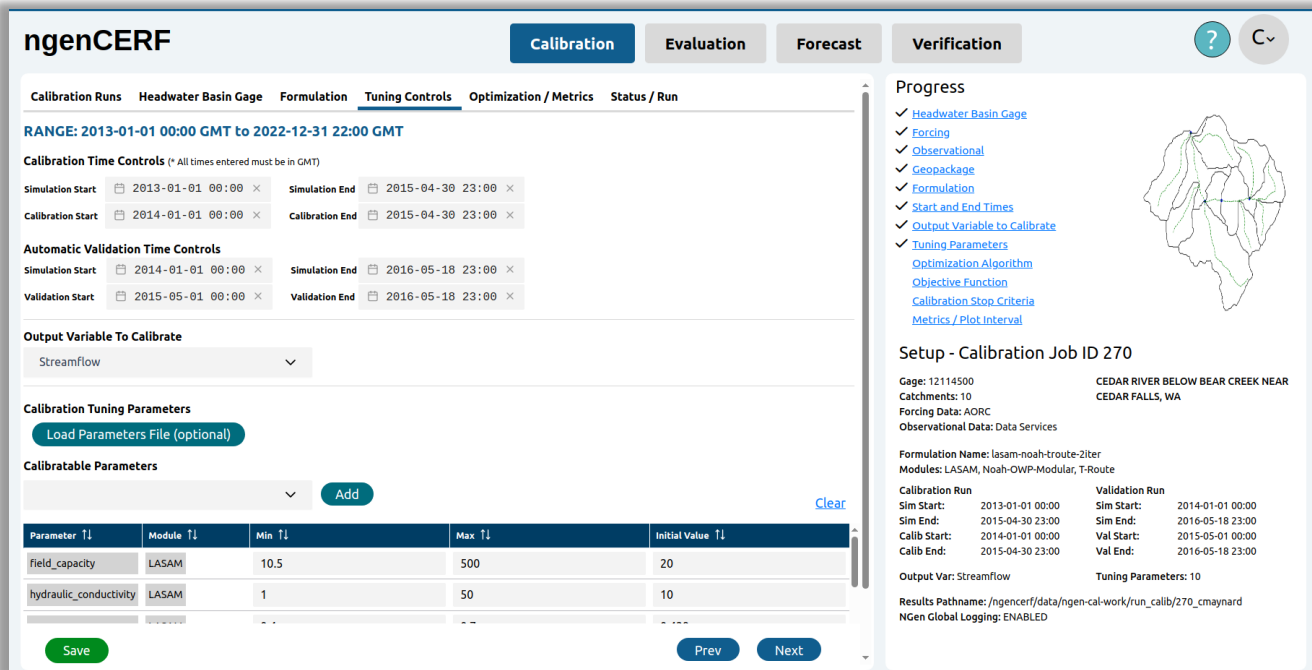
A calibration run requires four (4) different times be defined: the simulation start and end times and the calibration start and end times. The calibration start time provides the hydrologist the ability to run the simulation for a period to “spin up” the model states before starting the official calibration. This spinup period is the duration of time between the Simulation Start and the Calibration Start time for calibration and the Simulation Start and Validation Start time for validation.

A validation run is executed immediately following the calibration run but for a different time-period than the calibration run. The validation run will use the best parameters set as determined programmatically by the calibration run to validate the parameters against a different timeframe.

The validation run times are like the calibration run requiring four (4) different times: the simulation start and end times as well as the validation start and end times. The validation start time provides the hydrologist the ability to run the simulation for a period to “warmed up” the model states before starting the official validation.

The Range of Time available for both the Calibration and the Validation runs is constrained by the intersection of available times in the forcing data set and the observational data set.

The Calibration Tuning Parameters are module parameters used to fine tune the model through each iterative run of ngen. These calibratable parameters are based on the modules in the Formulation. They can be individually added using the pulldown or uploaded via a file. The parameters in this pulldown list are based on the modules selected in the Formulation.



User Actions:

Calibration Time Controls

Simulation Start

Date and hour to start the calibration simulation

Simulation End

Date and hour to stop the calibration simulation

Calibration Start

Date and hour to start calibrating.

Must be $\geq$ Simulation Start and $<$ Simulation End

Calibration End

Date and hour to stop calibration.

Must be $>$ Calibration Start and $\leq$ Simulation End

Validation Time Controls

Simulation Start

Date and hour to start the validation simulation

Must be outside of the Calibration Start and End period.

Simulation End

Date and hour to stop the validation simulation

Validation Start

Date and hour to start validation.

Must be $\geq$ Simulation Start and $<$ Simulation End

Validation End

Date and hour to stop validation.

Must be $>$ Validation Start and $\leq$ Simulation End

Output Variable to Calibrate³

Select an output variable from the pulldown menu. Currently Streamflow is the only support output for calibration.

Calibration Tuning Parameters³

Load Parameters

Optionally upload a file of preselected calibratable parameters associated with the formulation. This file can be either CSV, tab or space delimited and must contain the parameter name, module name, minimum value, maximum value, and initial value. See Figure 2 for an example CSV file.

This file is read and its contents added to the Parameter table where the parameters can be modified or deleted.

Add	Button to add the selected calibratable parameter from the pulldown menu. This menu is populated with calibratable parameters based on the modules in the formulation.
Parameter Table	After being added to the table, a calibratable parameter can be edited or removed.
Right Click	Right click and select Delete to remove a calibratable parameter from the table.
Save	Select this button to save the data entered on the tab. This can be done periodically or after making all the selections.
Revert	Select this button to reset selections for this tabbed work area.
Prev	Move to the Formulation Tab. If there is any unsaved data, the user will first be prompted to save.
Next	Move to the Optimization / Metrics Tab. If there is any unsaved data, the user will first be prompted to save

See the [BMI Exchange Items and Module Parameters Confluence Page](#) for a complete list of Calibratable Parameters per module and Section 6.4 for more details.

```
param,min,max,init,model
field_capacity,10.5,500.0,20.0,LASAM
hydraulic_conductivity,1.0,50.0,10.0,LASAM
smcmax,0.4,0.7,0.439,LASAM
```

Figure 2 Example Calibratable Parameters CSV file

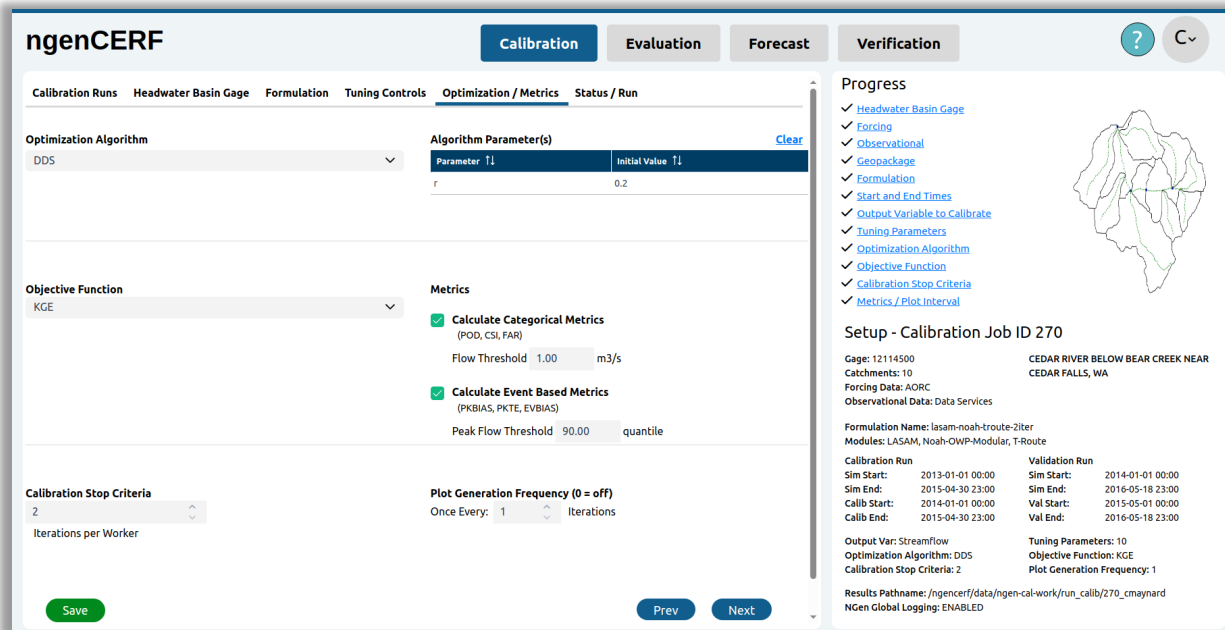
Important:

- *Tuning Controls cannot be selected until both the Headwater Basin Gage tab data and the Formulation are selected. If the Headwater Basin Gage is changed, the times entered for the Tuning controls will be cleared. If the Formulation is changed, the Calibration Tuning Parameters will be updated, removing any parameters for modules that were removed from the Formulation*
- *If the headwater basin gage, forcing data or observational data is changed, the times entered for the Calibration and Validation will be cleared.*
- *If the formulation of modules selected is changed, the Calibration Tuning Parameters and the Output Variable to Calibrate will be cleared.*

5.4.5 Optimization/Metrics Tab

The Optimization/Metrics tab³ provides the ability to specify inputs to the calibration manager that control the calibration behavior.

In this tab section, the user selects the Optimization Algorithm, the Objective Function, whether to calculate the Categorical Metrics and/or the Event Based Metrics and if selected, the associated dependent input, the Calibration Stop Criteria, and the Plot Generation Frequency.



ngenCERF

Calibration Runs | Headwater Basin Gage | Formulation | Tuning Controls | **Optimization / Metrics** | Status / Run

Optimization Algorithm
DDS

Algorithm Parameter(s)

Parameter	Initial Value
r	0.2

Objective Function
KGE

Metrics
☒ Calculate Categorical Metrics (POD, CSI, FAR)
 Flow Threshold: 1.00 m³/s
☒ Calculate Event Based Metrics (PKBIAS, PKTE, EVBIAS)
 Peak Flow Threshold: 90.00 quantile

Calibration Stop Criteria
2
Iterations per Worker

Plot Generation Frequency (0 = off)
Once Every: 1 Iterations

Progress
☒ Headwater Basin Gage
☒ Forcing
☒ Observational
☒ Geopackage
☒ Formulation
☒ Start and End Times
☒ Output Variable to Calibrate
☒ Tuning Parameters
☒ Optimization Algorithm
☒ Objective Function
☒ Calibration Stop Criteria
☒ Metrics / Plot Interval

Setup - Calibration Job ID 270
 Gage: 12114500 CEDAR RIVER BELOW BEAR CREEK NEAR
 Catchments: 10 CEDAR FALLS, WA
 Forcing Data: AORC
 Observational Data: Data Services
 Formulation Name: lasam-noah-trout-2iter
 Modules: LASAM, Noah-OVP-Modular, T-Route
 Calibration Run
 Sim Start: 2013-01-01 00:00
 Sim End: 2015-04-30 23:00
 Calib Start: 2014-01-01 00:00
 Calib End: 2015-04-30 23:00
 Validation Run
 Sim Start: 2014-01-01 00:00
 Sim End: 2016-05-18 23:00
 Val Start: 2015-05-01 00:00
 Val End: 2016-05-18 23:00
 Output Var: Streamflow
 Optimization Algorithm: DDS
 Calibration Stop Criteria: 2
 Tuning Parameters: 10
 Objective Function: KGE
 Plot Generation Frequency: 1
 Results Pathname: /ngenCERF/data/ngen-cal-work/run_calib/270_cmaynard
 NGen Global Logging: ENABLED

User Actions:

Optimization Algorithm

Select the optimization algorithm from the pulldown. Options are DDS, GWO, and PSO. DDS has one worker while GWO and PSO have multiple workers.

Algorithm Parameters

Based on the Optimization Algorithm chosen, this table will be auto populated, and the user may update the default values.

Objective Function

User selects a metric from the list of metrics in the pulldown. If the metric chosen is a Categorical Metric (POD, CSI, FAR) then the user must supply the Flow Threshold Value in m³/s units.

If the metric chosen is an Event Based Metric (PKBIAS, PKTE, EVBIAS) then the user must supply the Peak Flow Threshold Value in quantile units.

Flow Threshold	Required input in m^3/s units if a categorical metric is selected as the Objective Function.
Peak Flow Threshold	Required input in quartile units if an event-based metric is selected as the Objective Function.
Additional Metrics	Select additional metrics to be calculated. All other metrics are calculated by the calibration manager except Categorical and Event Based metrics.
Calculate Categorical Metrics	Checkbox to optionally calculate the categorical metrics POD, CSI and FAR. If checked the user must supply the dependent input. The user must supply the Flow Threshold.
Flow Threshold	Required input in m^3/s units if Calculate Categorical Metrics checkbox selected.
Calculate Event Based Metrics	Checkbox to optionally calculate the event-based metrics PKBIAS, PKTE, EVBIAS. The user must supply the Peak Flow Threshold.
Peak Flow Threshold	Required input in quartile units if Calculate Event Based Metrics checkbox is selected.
Calibration Stop Criteria	Select the number of Worker Iterations to execute before stopping the calibration.
Plot Generation Frequency	Number of iterations to run before generating a new set of plots that can be viewed by the user during the calibration run. These plots are cumulative and include data from all previous iterations.
Save	Select this button to save the data entered on the tab. This can be done periodically or after making all selections.
Revert	Select this button to reset selections for this tabbed work area.
Prev	Move to the Tuning Controls Tab. If there is any unsaved data, the user will first be prompted to save.
Next	Move to the Run / Status Tab. If there is any unsaved data, the user will first be prompted to save

Table 5: Calibration Metrics

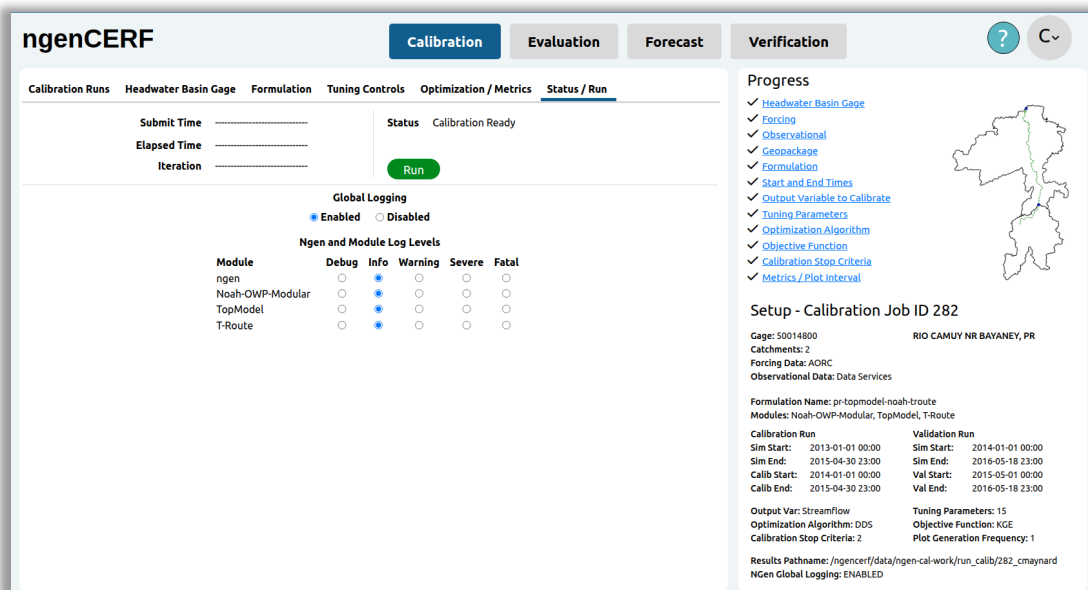
Metric Acronym	Long Description
COR	Pearson Correlation
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
RSR	Ratio of RMSE to standard deviation of observation
PBIAS	Percent Bias
KGE	Kling-Gupta Efficiency
NSE	Nash-Sutcliffe-Efficiency
LogNSE	NSE of logarithmic values
NNSE	Normalized NSE
POD	Probability of Detection (Categorical Metric)
CSI	Critical Success Index (Categorical Metric)
FAR	False Alarm Ratio (Categorical Metric)
HFDC	Percent bias of high flow segment of flow duration curve
LFDC	Percent bias of low flow segment of flow duration curve
PKBIAS	Absolute Peak Flow Bias (Event Based Metric)
PKTE	Peak Flow Timing Error (Event Based Metric)
EVBIAS	Event Volume Bias (Event Based Metric)
MFDC	Percent bias of middle flow segment of flow duration curve
FBIAS	Frequency bias
NSEwt	Weighted NSE (NSE and LogNSE)

5.4.6 Run / Status Tab

This tab is designed to allow the user to review their setup, start the calibration run and monitor its progress. It has two (2) distinct views.

View 1: Run

This view provides the ability for the user to review the setup on the right side of the screen, which has been updated throughout the user setup and update logging controls. After the user clicks the Run button, the Status view will be displayed.



The screenshot displays the ngenCERF software interface. The top navigation bar includes tabs for Calibration, Evaluation, Forecast, and Verification. The Calibration tab is active, and the 'Status / Run' sub-tab is selected. On the left, there are input fields for 'Submit Time', 'Elapsed Time', and 'Iteration', along with a 'Run' button. Below these, there are checkboxes for 'Global Logging' (Enabled/Disabled) and a table for 'Ngen and Module Log Levels' with columns for Debug, Info, Warning, Severe, and Fatal. The right side of the interface shows a 'Progress' section with a checklist of items like 'Headwater Basin Gage', 'Forcing', 'Observational', etc., and a map of a river network. Below the progress section is the 'Setup - Calibration Job ID 282' area, which contains detailed information about the calibration run, including the model name, catchments, forcing data, and various time parameters for simulation and validation.

User Actions:
Run

If the status is Calibration Ready, clicking this button starts the calibration run. If it is not ready, something is missing from the calibration setup.

Global Logging

Allows the user to Enable (default) or Disable logging in ngen and the formulation modules.

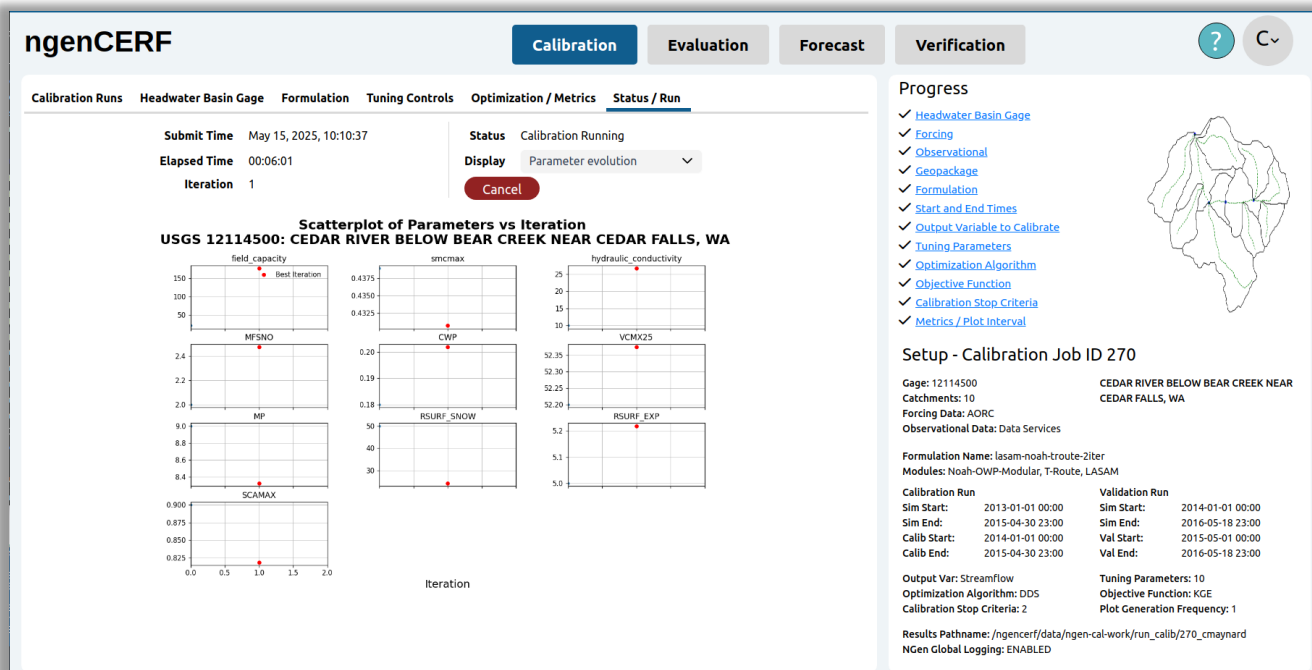
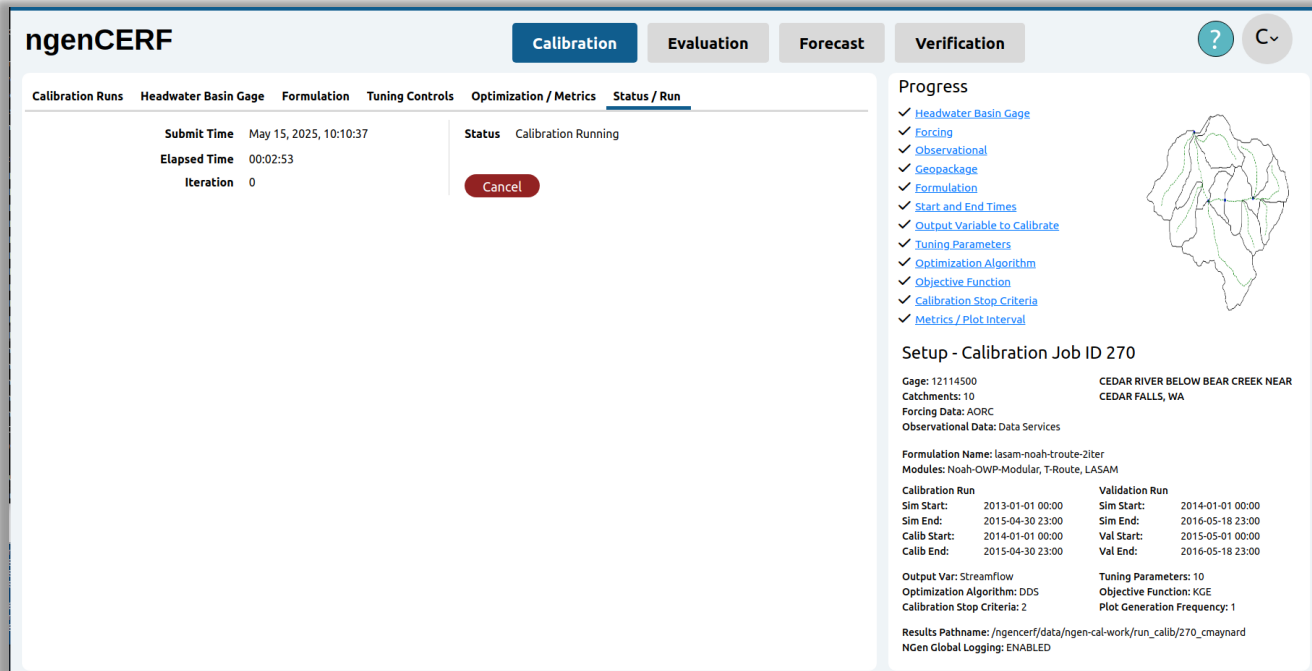
Log Levels

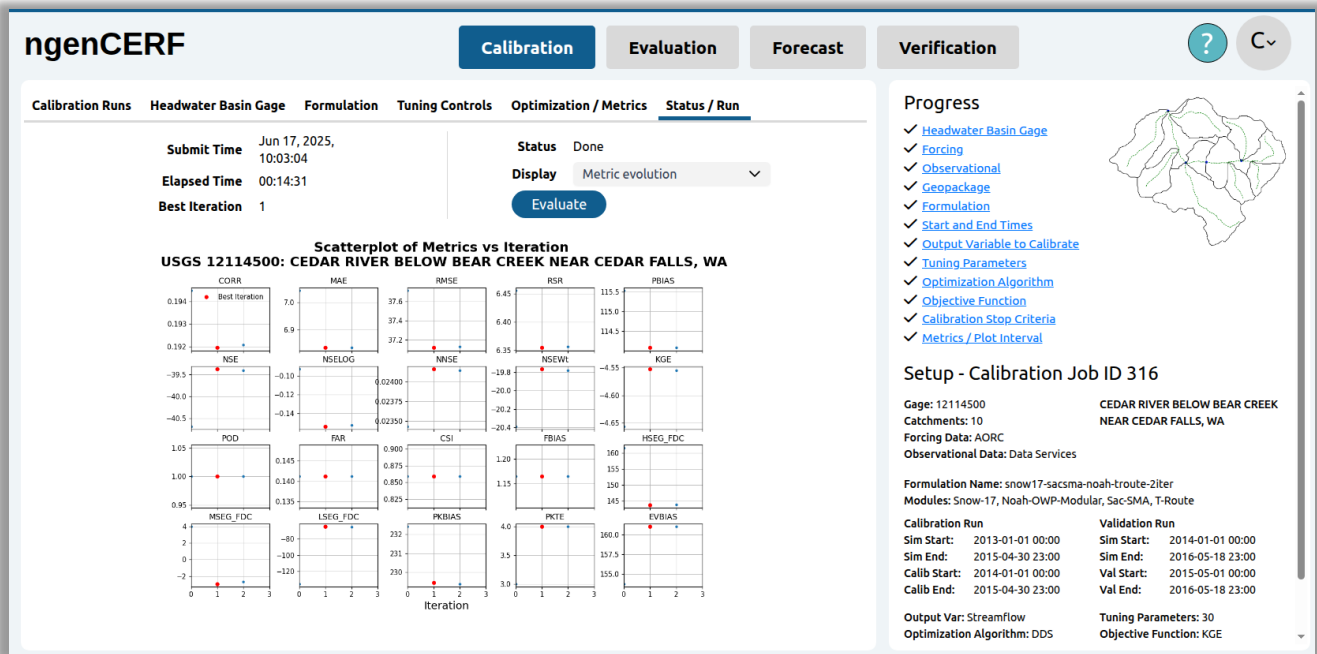
Select the log level for ngen and each module in the formulation. Default log level is INFO.

View 2: Status Area

After the user clicks Run, the calibration job data is prepared then the calibration will start. The status, job run time, and the current completed iteration are displayed and updated throughout the calibration run. Once a job starts running, the associated log files are viewable from the dropdown menu. After iteration 1 appears in the status area, the user can select a calibration plot to be displayed. (Iteration 0 is a control run and plots are not generated.) When the job status is Done, the user may look at validation plots or go to the Evaluation workflow to perform more evaluation of the job results, including running another validation on an alternate set of parameters. If a job fails for any reason, the log file is displayed in the status area.

Calibration Runs	Headwater Basin Gage	Formulation	Tuning Controls	Optimization / Metrics	Status / Run
Submit Time ----- Elapsed Time ----- Iteration -----			Status Validating and Preparing Job Data		
					





User Actions:

Display

Select a plot or log to display from the pulldown menu. These plots are updated at the Plot Generation Frequency and are cumulative. Logs are available once the job starts running. Calibration plots are available after iteration 0. Validation plots are available when the job is done.

Cancel

Cancel the currently running calibration. Depending on when this button is clicked, the Automatic Validation Run will not be executed.

Evaluate

When the job successfully completes, this button appears and takes the user to the Evaluation workflow where further evaluation and permutation comparison can be done.

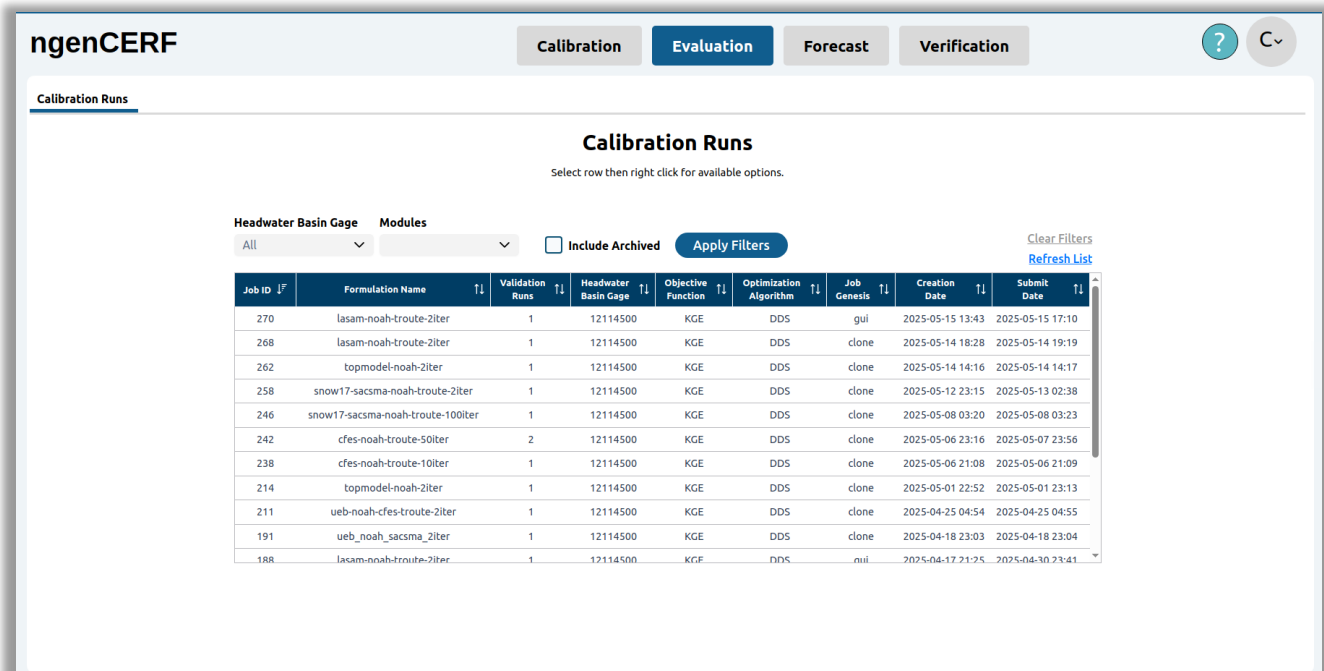
5.5 Evaluation Workflow

The Evaluation Interface provides the ability for the user to review the output from a previous calibration or validation run. Currently the calibration manager retains the data used and generated from the control run, last iteration and the best parameter set runs. After selecting the calibration run of interest, the user can view this data in the **Evaluate** tab. If the user wishes to view data from a different iteration, another validation must be executed by selecting a parameter set from the **Select Alternate Iteration** tab. This will re-generate the necessary data.

5.5.1 Calibration Runs Tab

The Calibration Runs tab provides the user the ability to select which calibration job to evaluate. There are alternate views of this tab if the user must specify which validation run or which permutations (calibration jobs) to compare.

Evaluation Calibration Runs Landing View:



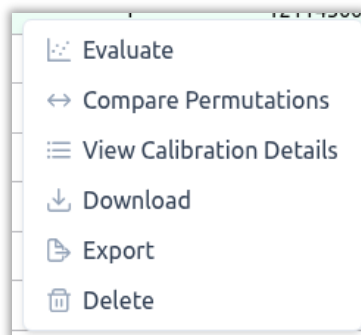
ngenCERF Calibration Evaluation Forecast Verification

Calibration Runs
Select row then right click for available options.

Headwater Basin Gage: All Modules: ☐ Include Archived [Clear Filters](#) [Refresh List](#)

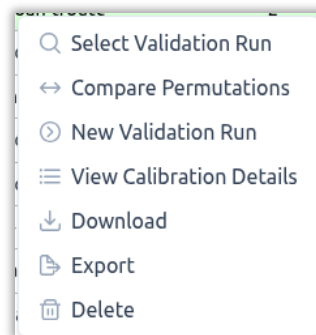
Job ID	Formulation Name	Validation Runs	Headwater Basin Gage	Objective Function	Optimization Algorithm	Job Genesis	Creation Date	Submit Date
270	lasam-noah-troute-2iter	1	12114500	KGE	DDS	gui	2025-05-15 13:43	2025-05-15 17:10
268	lasam-noah-troute-2iter	1	12114500	KGE	DDS	clone	2025-05-14 18:28	2025-05-14 19:19
262	topmodel-noah-2iter	1	12114500	KGE	DDS	clone	2025-05-14 14:16	2025-05-14 14:17
258	snow17-sacma-noah-troute-2iter	1	12114500	KGE	DDS	clone	2025-05-12 23:15	2025-05-13 02:38
246	snow17-sacma-noah-troute-100iter	1	12114500	KGE	DDS	clone	2025-05-08 03:20	2025-05-08 03:23
242	cfes-noah-troute-50iter	2	12114500	KGE	DDS	clone	2025-05-06 23:16	2025-05-07 23:56
238	cfes-noah-troute-10iter	1	12114500	KGE	DDS	clone	2025-05-06 21:08	2025-05-06 21:09
214	topmodel-noah-2iter	1	12114500	KGE	DDS	clone	2025-05-01 22:52	2025-05-01 23:13
211	ueb-noah-cfes-troute-2iter	1	12114500	KGE	DDS	clone	2025-04-25 04:54	2025-04-25 04:55
191	ueb_noah_sacma_2iter	1	12114500	KGE	DDS	clone	2025-04-18 23:03	2025-04-18 23:04
188	lasam-noah-troute-2iter	1	12114500	KGE	DDS	gui	2025-04-17 21:25	2025-04-30 23:41

Right Click Menu: 1 Validation Run:



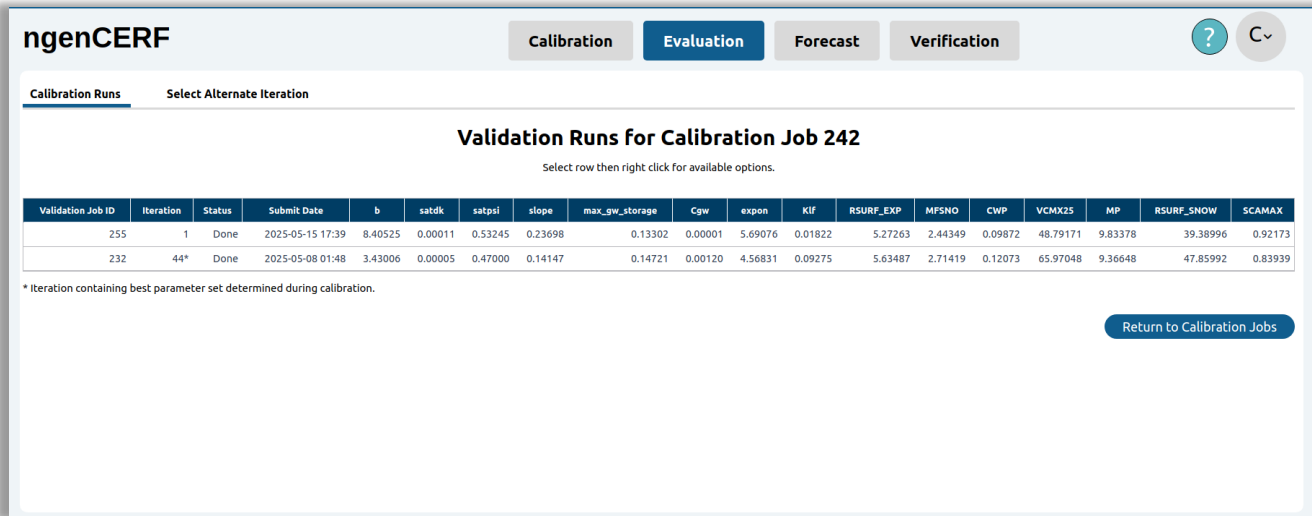
- Evaluate
- Compare Permutations
- View Calibration Details
- Download
- Export
- Delete

Right Click Menu: Multiple Validation Runs:



- Select Validation Run
- Compare Permutations
- New Validation Run
- View Calibration Details
- Download
- Export
- Delete

Evaluation Calibration Runs Select Validation Run View:



ngenCERF Calibration Evaluation Forecast Verification

Calibration Runs Select Alternate Iteration

Validation Runs for Calibration Job 242

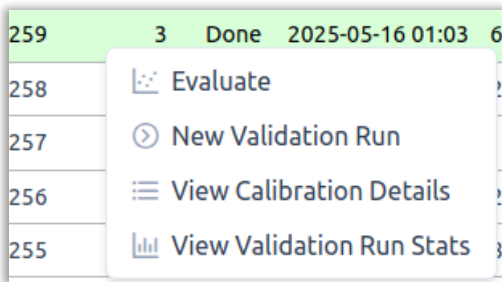
Select row then right click for available options.

Validation Job ID	Iteration	Status	Submit Date	b	satdk	satpsi	slope	max_gw_storage	Cgw	expon	Klf	RSURF_EXP	MFSNO	CWP	VCMX25	MP	RSURF_SNOW	SCAMAX	
255	1	Done	2025-05-15 17:39	8.40525	0.00011	0.53245	0.23698		0.13302	0.00001	5.69076	0.01822	5.27263	2.44349	0.09872	48.79171	9.83378	39.38996	0.92173
232	44*	Done	2025-05-08 01:48	3.43006	0.00005	0.47000	0.14147		0.14721	0.00120	4.56831	0.09275	5.63487	2.71419	0.12073	65.97048	9.36648	47.85992	0.83939

* Iteration containing best parameter set determined during calibration.

[Return to Calibration Jobs](#)

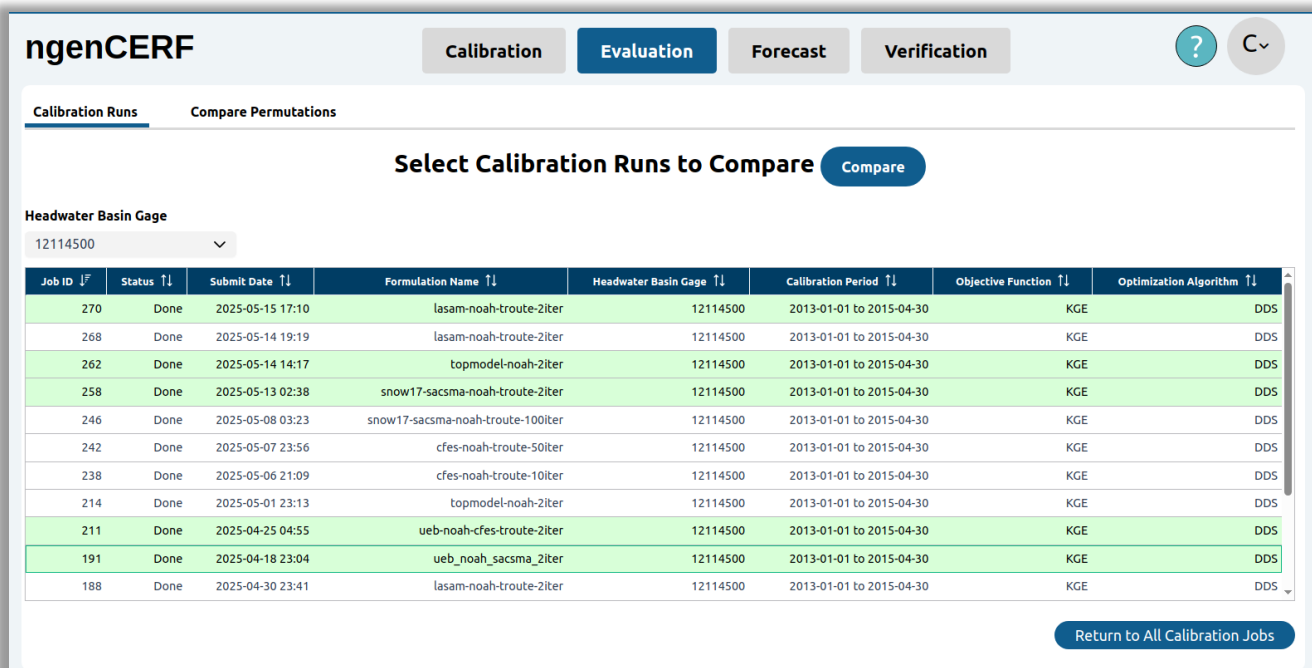
Right Click on Selected Validation Run:



259 3 Done 2025-05-16 01:03 6

- Evaluate
- New Validation Run
- View Calibration Details
- View Validation Run Stats

Compare Permutations Select Calibrations View:



ngenCERF Calibration Evaluation Forecast Verification

Calibration Runs Compare Permutations

Select Calibration Runs to Compare

[Compare](#)

Headwater Basin Gage
12114500

Job ID	Status	Submit Date	Formulation Name	Headwater Basin Gage	Calibration Period	Objective Function	Optimization Algorithm
270	Done	2025-05-15 17:10	lasam-noah-troute-2liter	12114500	2013-01-01 to 2015-04-30	KGE	DDS
268	Done	2025-05-14 19:19	lasam-noah-troute-2liter	12114500	2013-01-01 to 2015-04-30	KGE	DDS
262	Done	2025-05-14 14:17	topmodel-noah-2liter	12114500	2013-01-01 to 2015-04-30	KGE	DDS
258	Done	2025-05-13 02:38	snow17-sacsma-noah-troute-2liter	12114500	2013-01-01 to 2015-04-30	KGE	DDS
246	Done	2025-05-08 03:23	snow17-sacsma-noah-troute-100liter	12114500	2013-01-01 to 2015-04-30	KGE	DDS
242	Done	2025-05-07 23:56	cfes-noah-troute-50liter	12114500	2013-01-01 to 2015-04-30	KGE	DDS
238	Done	2025-05-06 21:09	cfes-noah-troute-10liter	12114500	2013-01-01 to 2015-04-30	KGE	DDS
214	Done	2025-05-01 23:13	topmodel-noah-2liter	12114500	2013-01-01 to 2015-04-30	KGE	DDS
211	Done	2025-04-25 04:55	ueb-noah-cfes-troute-2liter	12114500	2013-01-01 to 2015-04-30	KGE	DDS
191	Done	2025-04-18 23:04	ueb_noah_sacsma_2liter	12114500	2013-01-01 to 2015-04-30	KGE	DDS
188	Done	2025-04-30 23:41	lasam-noah-troute-2liter	12114500	2013-01-01 to 2015-04-30	KGE	DDS

[Return to All Calibration Jobs](#)

User Actions:

Filters

Headwater Basin Gage

Use this to filter the Calibration Runs table contents by a headwater basin gage ID.

Filters

Filter by Headwater Basin Gage, multi-select status, multi-select modules, and include or exclude archived jobs. Must click Apply Filters to filter the table. The Multi-select filters will show any job that matches on of the selected items. i.e. It is an “or” filter not an “and” filter.

Right Click

Evaluate

Select from the following options:

Takes user to the Evaluate tab. Appears if calibration job has 0 or 1 validation jobs.

Select Validation Run

When there is more than one (1) validation job, this will replace the Evaluate option. Selecting this will take the user to the view showing the table of associated validation jobs.

Compare Permutations

Selecting this will take the user to the view showing the table of validation jobs with the same gage ID.

New Validation Run

After a calibration run has been selected this button will appear. Selecting this button will take the user to the Select Alternate Iteration to select the parameter set to use for another validation run.

View Calibration Details

View the summary details of the calibration job.

View Validation Run Status

View the run status of a validation run.

Download Calibration Data

Download all data associated with a Done calibration job.

Export

Export the Calibration Definition File. (Used by the CLI.)

Delete Calibration Job

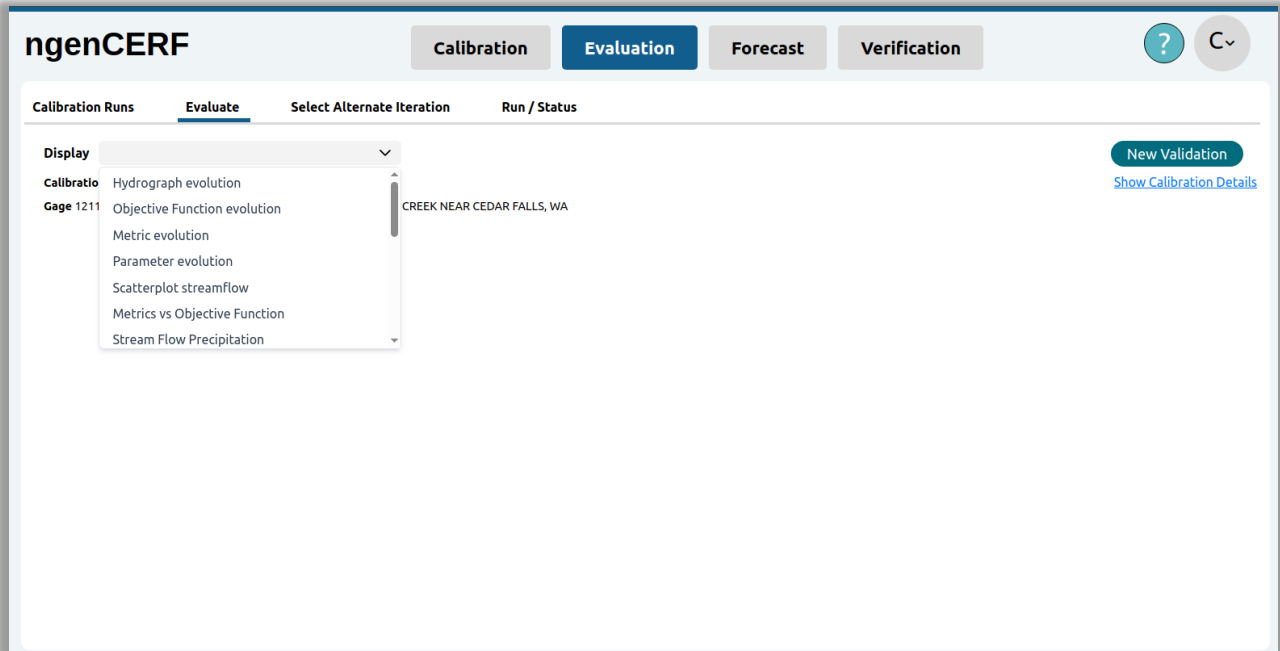
Permanently deletes the selected calibration job(s) from the system.

Return To All Calibration Jobs

When on the view to select a Validation Run or select Calibration Jobs to compare, click this button to return to the table of Calibration Runs. Clicking the Calibrations Runs tab name will not work in these views.

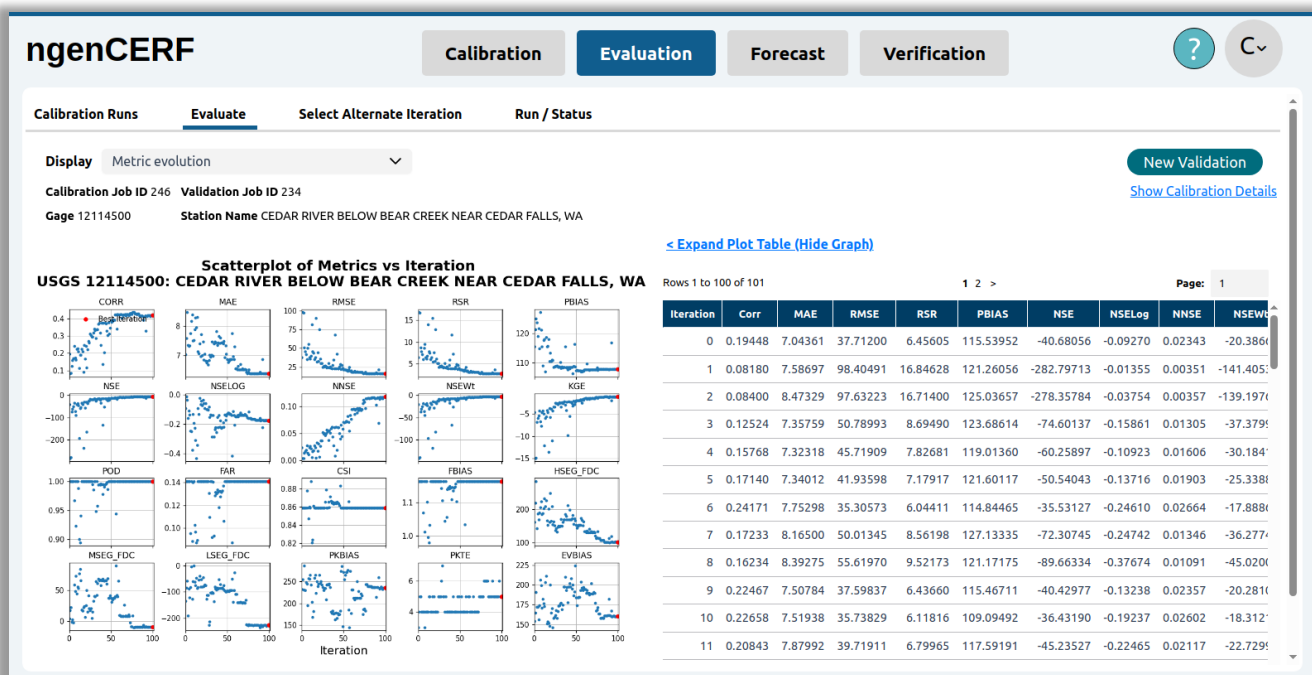
5.5.2 Evaluate Tab

The user is provided a dropdown to select plots generated during calibration and validation plots generated during the calibration as well as Time Series Viewer.



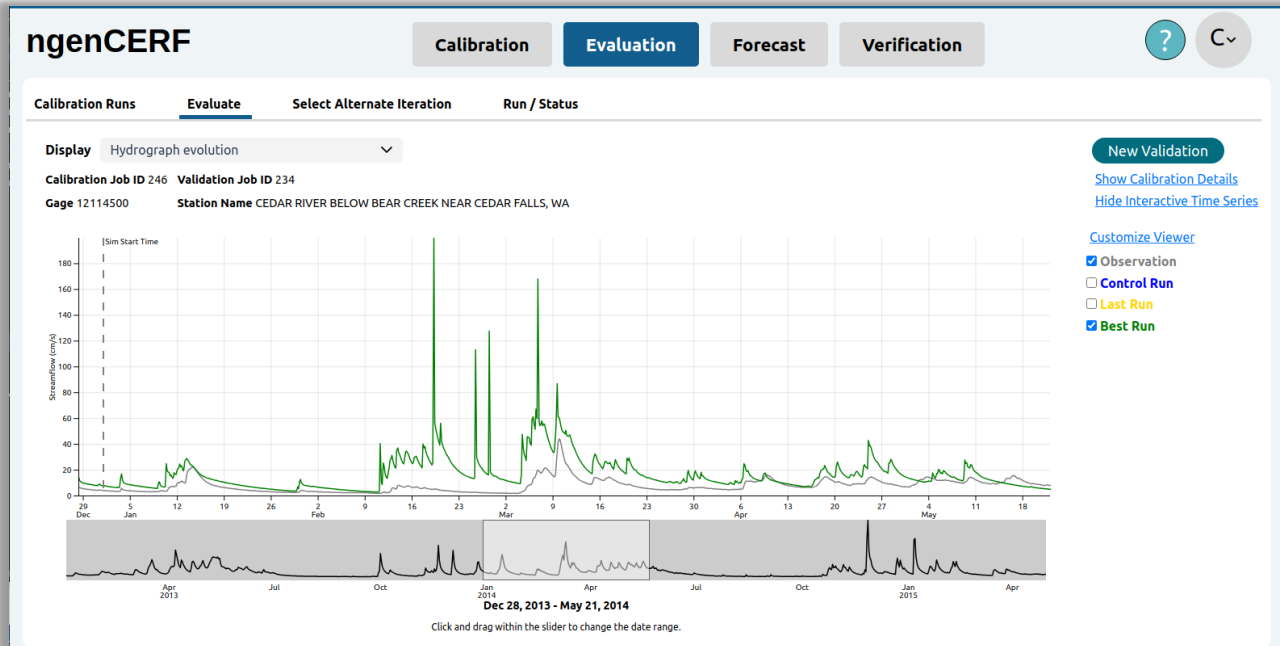
5.5.2.1 Plots and Tables

The calibration manager program creates plot and saves iteration data. These are available for viewing side-by-side or in table form. For example, **Metric Evolution**.



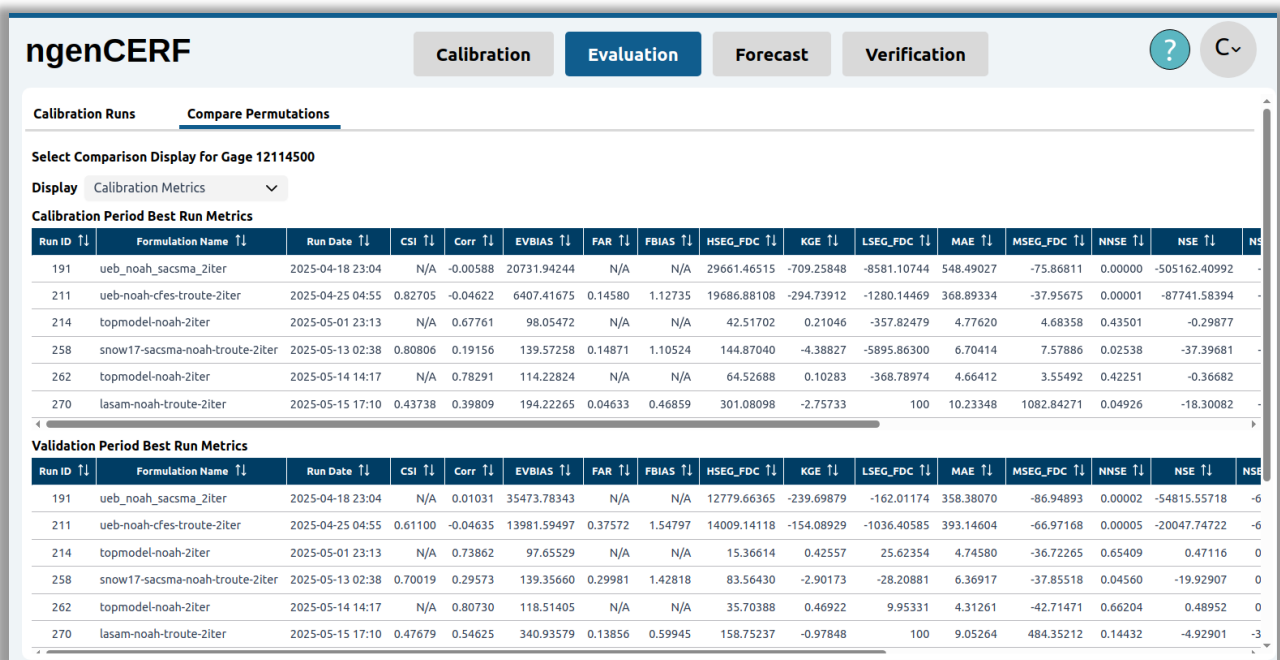
5.5.2.2 Time Series Viewer

A time series viewer is available for plots with saved time series data, such as Hydrograph Evolution.



5.5.3 Compare Permutations View

This tab allows the user to make a comparison of different calibration permutation runs.



Run ID	Formulation Name	Run Date	CSI	Corr	EVBIAS	FAR	FBIAS	HSEG_FDC	KGE	LSEG_FDC	MAE	MSEG_FDC	NNSE	NSE	NSI
191	ueb_noah_sacma_2iter	2025-04-18 23:04	N/A	-0.00588	20731.94244	N/A	N/A	29661.46515	-709.25848	-8581.10744	548.49027	-75.86811	0.00000	-505162.40992	-
211	ueb-noah-cfes-troute-2iter	2025-04-25 04:55	0.82705	-0.04622	6407.41675	0.14580	1.12735	19686.88108	-294.73912	-1280.14469	368.89334	-37.95675	0.00001	-87741.58394	-
214	topmodel-noah-2iter	2025-05-01 23:13	N/A	0.67761	98.05472	N/A	N/A	42.51702	0.21046	-357.82479	4.77620	4.68358	0.43501	-0.29877	-
258	snow17-sacma-noah-troute-2iter	2025-05-13 02:38	0.80806	0.19156	139.57258	0.14871	1.10524	144.87040	-4.38827	-5895.86300	6.70414	7.57886	0.02538	-37.39681	-
262	topmodel-noah-2iter	2025-05-14 14:17	N/A	0.78291	114.22824	N/A	N/A	64.52688	0.10283	-368.78974	4.66412	3.55492	0.42251	-0.36682	-
270	lasam-noah-troute-2iter	2025-05-15 17:10	0.43738	0.39809	194.22265	0.04633	0.46859	301.08098	-2.75733	100	10.23348	1082.84271	0.04926	-18.30082	-

Run ID	Formulation Name	Run Date	CSI	Corr	EVBIAS	FAR	FBIAS	HSEG_FDC	KGE	LSEG_FDC	MAE	MSEG_FDC	NNSE	NSE	NSI
191	ueb_noah_sacma_2iter	2025-04-18 23:04	N/A	0.01031	35473.78343	N/A	N/A	12779.66365	-239.69879	-162.01174	358.38070	-86.94893	0.00002	-54815.55718	-6
211	ueb-noah-cfes-troute-2iter	2025-04-25 04:55	0.61100	-0.04635	13981.59497	0.37572	1.54797	14009.14118	-154.08929	-1036.40585	393.14604	-66.97168	0.00005	-20047.74722	-6
214	topmodel-noah-2iter	2025-05-01 23:13	N/A	0.73862	97.65529	N/A	N/A	15.36614	0.42557	25.62354	4.74580	-36.72265	0.65409	0.47116	0
258	snow17-sacma-noah-troute-2iter	2025-05-13 02:38	0.70019	0.29573	139.35660	0.29981	1.42818	83.56430	-2.90173	-28.20881	6.36917	-37.85518	0.04560	-19.92907	0
262	topmodel-noah-2iter	2025-05-14 14:17	N/A	0.80730	118.51405	N/A	N/A	35.70388	0.46922	9.95331	4.31261	-42.71471	0.66204	0.48952	0
270	lasam-noah-troute-2iter	2025-05-15 17:10	0.47679	0.54625	340.93579	0.13856	0.59945	158.75237	-0.97848	100	9.05264	484.35212	0.14432	-4.92901	-3

5.5.4 Select Alternative Iteration Tab

This tab is activated when a New Validation Run action is selected. It allows the user to simultaneously view in two (2) separate tables the iteration metrics, including the objective function, and the associated set of parameters used for the iteration resulting in those metrics. To select the alternate set of parameters to use for a validation run, the user can select the iteration number from either table. After selected the user clicks **Next**.

ngenCERF

Calibration

Evaluation

Forecast

Verification

?

Cv

Calibration Runs

Select Alternate Iteration

Run / Status

Run Details - Calibration Job ID 242

Worker ↑↓	Iteration ↑↓	Validation Job ID ↑↓	Corr ↑↓	MAE ↑↓	RMSE ↑↓	RSR ↑↓	PBIAS ↑↓	NSE ↑↓	NSELog ↑↓	NNSE ↑↓	NSEwt ↑↓	KGE* ↑↓	POD ↑↓	FAR ↑↓	CSI ↑↓	FBIAS ↑↓	HSEC_FDC
NWM 3.0			0.8223	1.9059	6.5660	0.6015	-20.5042	0.6382	0.7970	0.7343	0.7176	0.7285	0.9954	0.0656	0.9304	1.0652	-15.6381
2lh26d9l	44 (Best)	232	0.6387	3.5767	5.5914	0.9572	15.7883	0.0838	-1.6260	0.5219	-0.7711	0.5634	0.8219	0.0001	0.8219	0.8220	24.3199
2lh26d9l	18		0.6003	3.7145	5.8567	1.0026	20.7466	-0.0053	-1.4693	0.4987	-0.7373	0.5183	0.8284	0.0001	0.8284	0.8285	23.3318
2lh26d9l	19		0.6244	3.6720	5.7282	0.9806	24.3206	0.0384	-1.1817	0.5098	-0.5717	0.5216	0.8678	0.0001	0.8677	0.8679	23.2453
2lh26d9l	20		0.5432	3.8580	6.1508	1.0530	19.8223	-0.1088	-1.5905	0.4742	-0.8496	0.4801	0.8290	0.0001	0.8290	0.8291	21.0112
2lh26d9l	21		0.5988	3.7534	5.8969	1.0095	16.1401	-0.0191	-1.7798	0.4953	-0.8995	0.5263	0.8188	0.0001	0.8188	0.8189	25.5294

* Metric used as Objective Function

Corresponding Calibration Tuning Parameters

Worker ↑↓	Iteration ↑↓	Validation Job ID ↑↓	b ↑↓	satdk ↑↓	satpsi ↑↓	slope ↑↓	max_gw_storage ↑↓	Cgw ↑↓	expon ↑↓	Klf ↑↓	RSURF_EXP ↑↓	MFSNO ↑↓	CWP ↑↓	VCMX25 ↑↓	MP ↑↓	RSURF
2lh26d9l	44 (Best)	232	3.4301	0.0000	0.4700	0.1415	0.1472	0.0012	4.5683	0.0927	5.6349	2.7142	0.1207	65.9705	9.3665	47.85
2lh26d9l	18		3.4323	0.0000	0.4700	0.1415	0.1128	0.0013	2.5355	0.1197	5.8081	2.9026	0.1728	47.2526	9.3665	56.00
2lh26d9l	19		2.4754	0.0000	0.4700	0.1415	0.1128	0.0013	3.2306	0.1197	5.8081	2.9026	0.1811	47.2526	9.3665	46.04
2lh26d9l	20		3.4301	0.0000	0.5005	0.1415	0.1128	0.0013	5.2924	0.2271	5.8081	2.9026	0.1547	47.2526	9.3665	38.94
2lh26d9l	21		3.4301	0.0000	0.4700	0.1415	0.1472	0.0013	5.3722	0.1197	5.8081	2.7142	0.0972	47.2526	9.3665	47.85

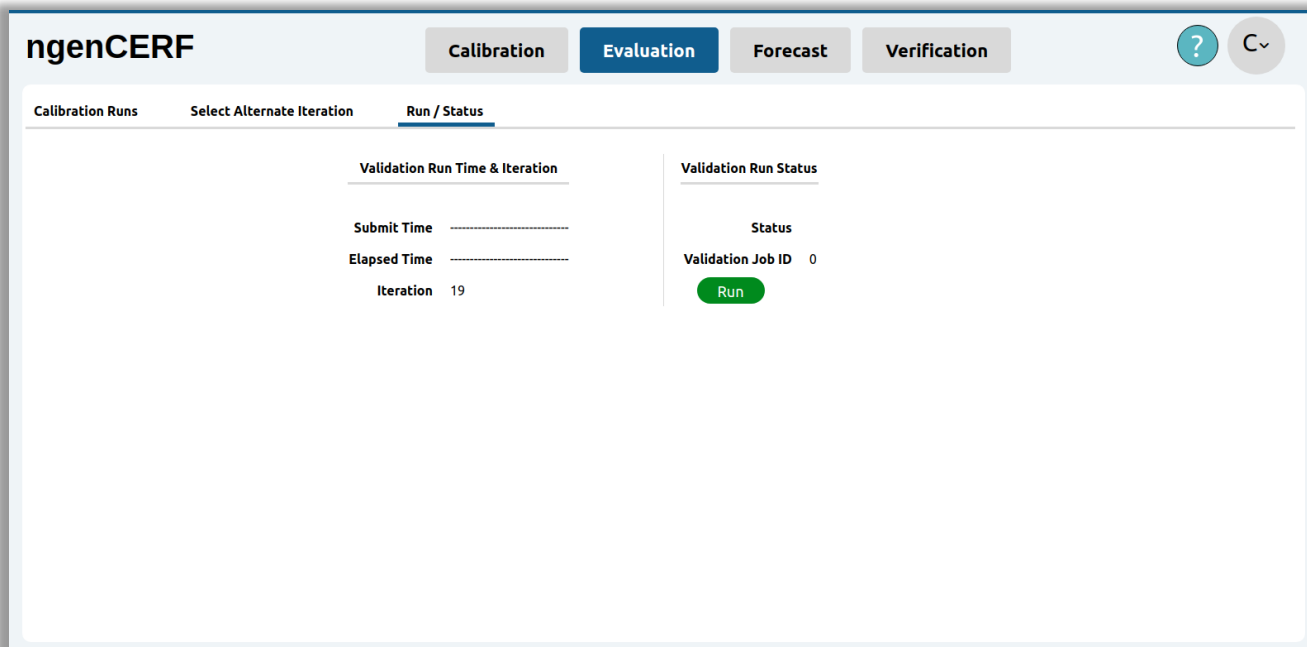
Next

Note:

- Run Details Table is column-sortable table with selectable entries
- Corresponding Calibration Tuning Parameters Table is column-sortable table with selectable entries

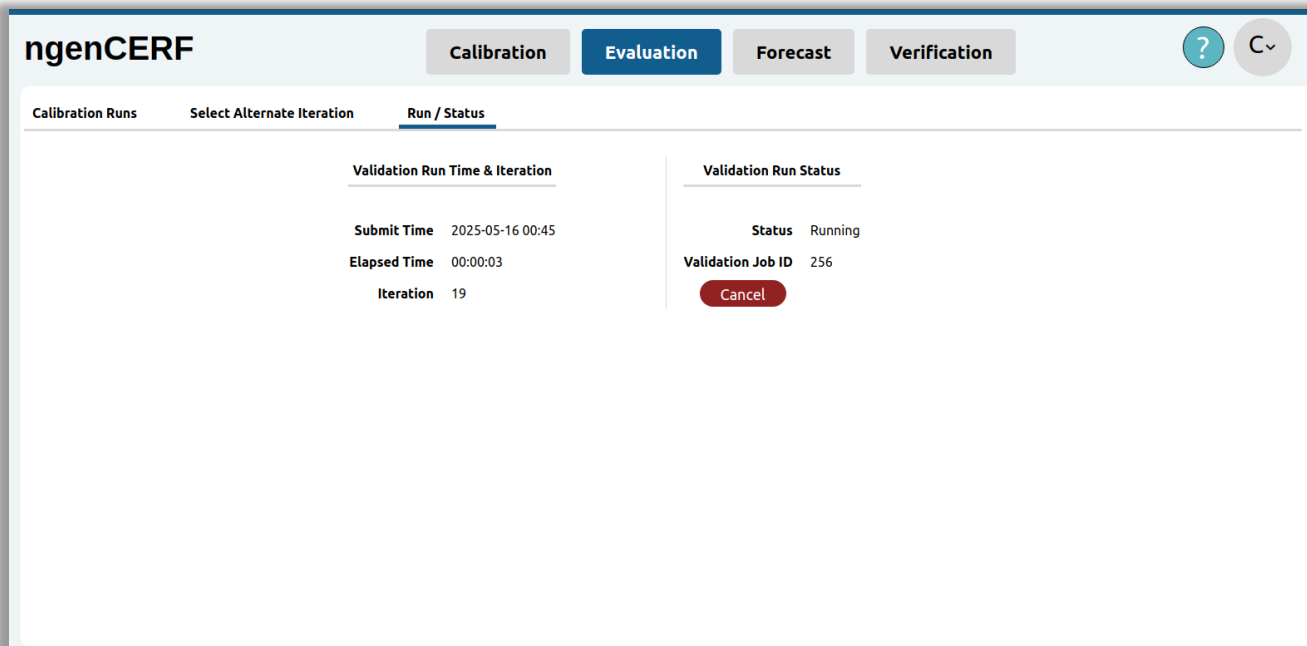
5.5.5 Run / Status Tab

This tab allows the user to start the validation run and, once running, provides the status of the run. When complete, the **Evaluate** appears. Click this to view the results of the validation run.



The screenshot shows the ngenCERF interface with the 'Evaluation' tab selected. The 'Run / Status' sub-tab is active. On the left, under 'Validation Run Time & Iteration', the 'Iteration' is 19. On the right, under 'Validation Run Status', the 'Status' is empty and the 'Validation Job ID' is 0. A green 'Run' button is visible.

Validation Run Time & Iteration		Validation Run Status	
Submit Time	-----	Status	
Elapsed Time	-----	Validation Job ID	0
Iteration	19		Run



The screenshot shows the ngenCERF interface with the 'Evaluation' tab selected. The 'Run / Status' sub-tab is active. On the left, under 'Validation Run Time & Iteration', the 'Submit Time' is 2025-05-16 00:45, 'Elapsed Time' is 00:00:03, and 'Iteration' is 19. On the right, under 'Validation Run Status', the 'Status' is 'Running' and the 'Validation Job ID' is 256. A red 'Cancel' button is visible.

Validation Run Time & Iteration		Validation Run Status	
Submit Time	2025-05-16 00:45	Status	Running
Elapsed Time	00:00:03	Validation Job ID	256
Iteration	19		Cancel

ngenCERF

CalibrationEvaluationForecastVerification

Calibration RunsEvaluateSelect Alternate IterationRun / Status

Validation Run Time & Iteration

Submit Time

2025-05-16 01:03

Elapsed Time

00:02:30

Iteration

3

Validation Run Status

Status

Done

Validation Job ID

259

Evaluate

User Actions:

Run

Cancel

Evaluate

Click to start the Validation Run

Click to cancel the Validation Run

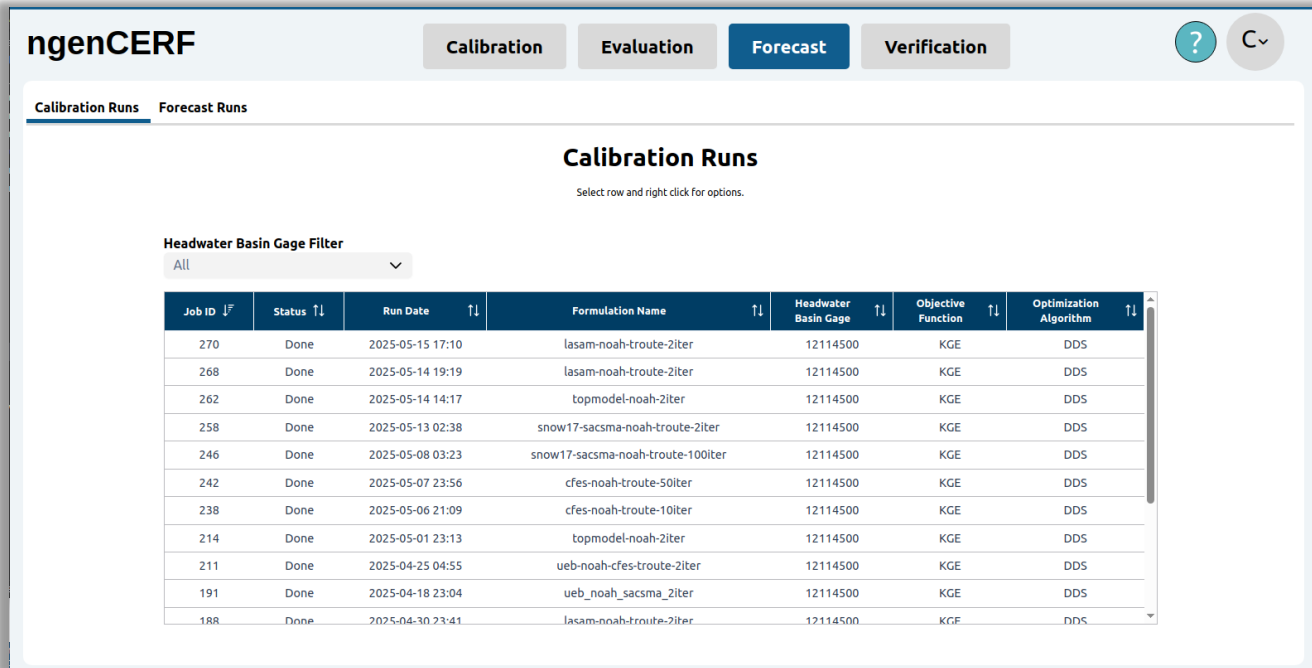
Go to the Evaluate tab

5.6 Forecast Workflow

The Forecast workflow provides the ability for the user to run a forecast using the best set of parameters from a previous calibration run.

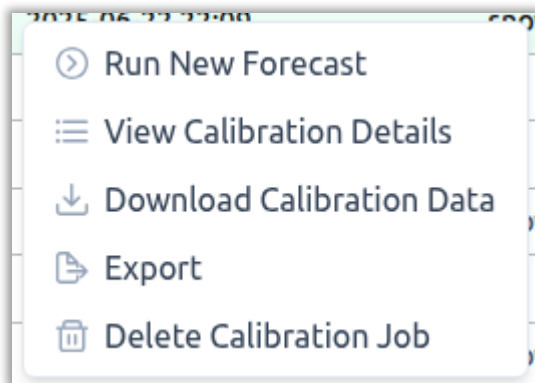
5.6.1 Calibration Runs Tab

The Calibration Runs tab provides the user the ability to select which calibration job to use as the basis for a forecast. After a calibration job is opened by double clicking or right clicking and selecting open, they will be moved to the Setup Forecast tab.



Job ID	Status	Run Date	Formulation Name	Headwater Basin Gage	Objective Function	Optimization Algorithm
270	Done	2025-05-15 17:10	lasam-noah-troute-2lter	12114500	KGE	DDS
268	Done	2025-05-14 19:19	lasam-noah-troute-2lter	12114500	KGE	DDS
262	Done	2025-05-14 14:17	topmodel-noah-2lter	12114500	KGE	DDS
258	Done	2025-05-13 02:38	snow17-sacsma-noah-troute-2lter	12114500	KGE	DDS
246	Done	2025-05-08 03:23	snow17-sacsma-noah-troute-100lter	12114500	KGE	DDS
242	Done	2025-05-07 23:56	cfes-noah-troute-50lter	12114500	KGE	DDS
238	Done	2025-05-06 21:09	cfes-noah-troute-10lter	12114500	KGE	DDS
214	Done	2025-05-01 23:13	topmodel-noah-2lter	12114500	KGE	DDS
211	Done	2025-04-25 04:55	ueb-noah-cfes-troute-2lter	12114500	KGE	DDS
191	Done	2025-04-18 23:04	ueb_noah_sacsma_2lter	12114500	KGE	DDS
188	Done	2025-04-30 23:41	lasam-noah-troute-2lter	12114500	KGE	DDS

Forecast Calibration Runs Context Menu:



User Actions:

Headwater Basin Gage Filter

Use this to filter the table contents by a headwater basin gage ID.

Calibration Runs Table

Select a calibration run to use as a basis for a forecast. The table is column sortable.

Right Click

Run New Forecast	Run a new forecast based on the selected calibration job.
View Calibration Details	View the summary details of the calibration job.
Download Calibration Data	Download all data associated with a Done calibration job.
Export	Export the Calibration Definition File. (Used by the CLI.)
Delete Calibration Job	Permanently deletes the selected calibration job(s) from the system.

5.6.2 Setup Forecast Tab

Use this tab to select the type of forecast cycle to run.

ngenCERF

Calibration

Evaluation

Forecast

Verification

?

Cv

Calibration Runs

Forecast Runs

Setup Forecast

Gage ID: 12114500

Calibration Job ID: 351

Forecast Configuration Selection

Select a configuration then click Next.

Configuration	Data Sources	Time Range (ngenCERF)
Analysis and Assimilation (AnA)	HRRR, RAP, MRMS-MS, MRMS-RO, USGS gages	3 hr
Short Range Forecast	HRRR, RAP	Latest forecast cycle, 18 hours
Extended AnA	RAP, HRRR, Stage IV	tbd
Medium Range Forecast	tbd	tbd
Long Range AnA	HRRR, RAP, MRMS-MS, MRMS-RO, USGS gages	tbd
Long Range Forecast	long_range_forecast	tbd

Configurations in grey are unavailable.

IMPORTANT

Short Range Forecast is for CONUS gages only.

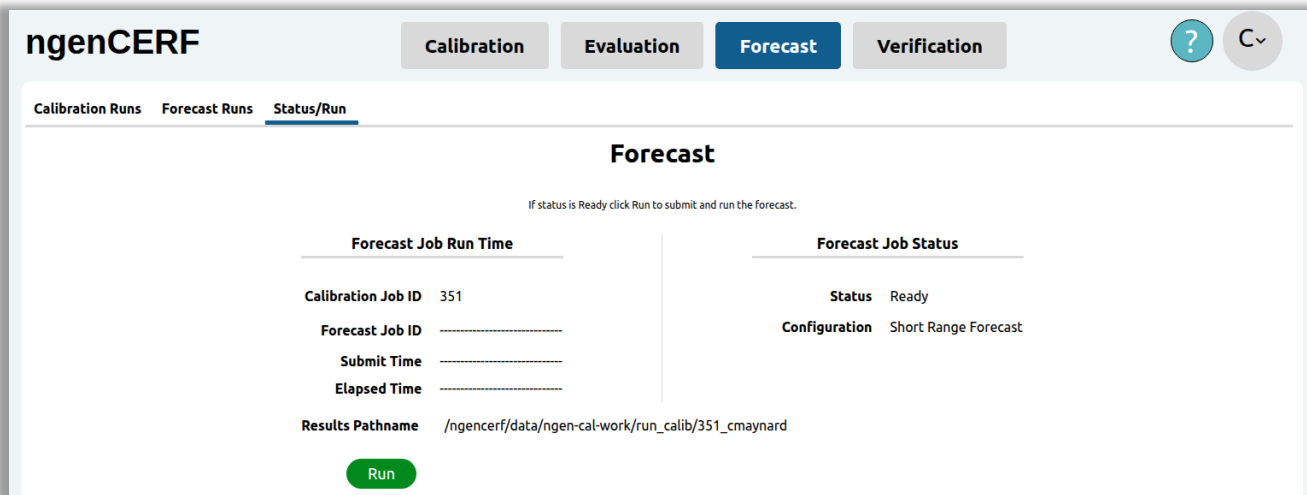
Use the ngenCERF CLI for oCONUS Forecasts.

Using an oCONUS gage will result in a Failed Forecast.

Next

5.6.3 Status / Run Tab

Use this tab to review the forecast setup and click Run to start the forecast.



ngenCERF Calibration Evaluation **Forecast** Verification ? C

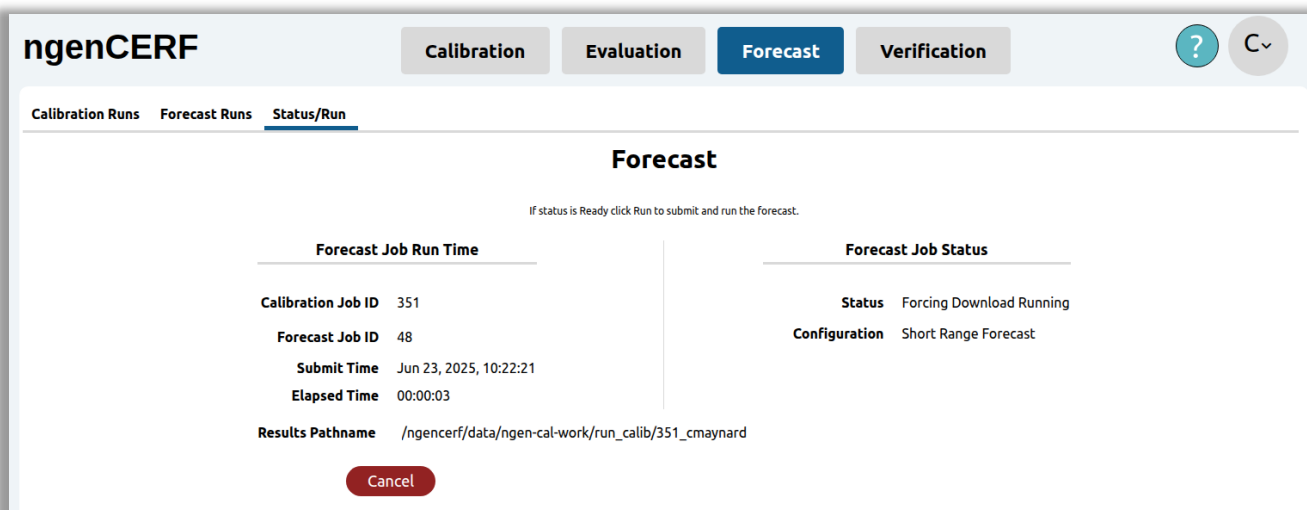
Calibration Runs Forecast Runs **Status/Run**

Forecast

If status is Ready click Run to submit and run the forecast.

Forecast Job Run Time		Forecast Job Status	
Calibration Job ID	351	Status	Ready
Forecast Job ID	-----	Configuration	Short Range Forecast
Submit Time	-----		
Elapsed Time	-----		
Results Pathname	/ngencerf/data/ngen-cal-work/run_calib/351_cmaynard		

Run



ngenCERF Calibration Evaluation **Forecast** Verification ? C

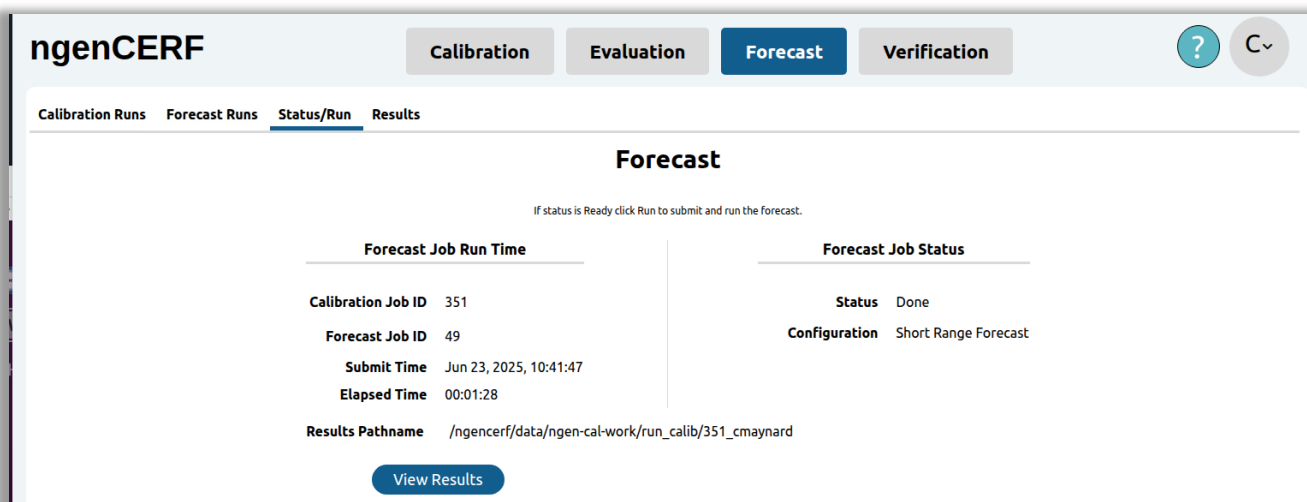
Calibration Runs Forecast Runs **Status/Run**

Forecast

If status is Ready click Run to submit and run the forecast.

Forecast Job Run Time		Forecast Job Status	
Calibration Job ID	351	Status	Forcing Download Running
Forecast Job ID	48	Configuration	Short Range Forecast
Submit Time	Jun 23, 2025, 10:22:21		
Elapsed Time	00:00:03		
Results Pathname	/ngencerf/data/ngen-cal-work/run_calib/351_cmaynard		

Cancel



ngenCERF Calibration Evaluation **Forecast** Verification ? C

Calibration Runs Forecast Runs **Status/Run** Results

Forecast

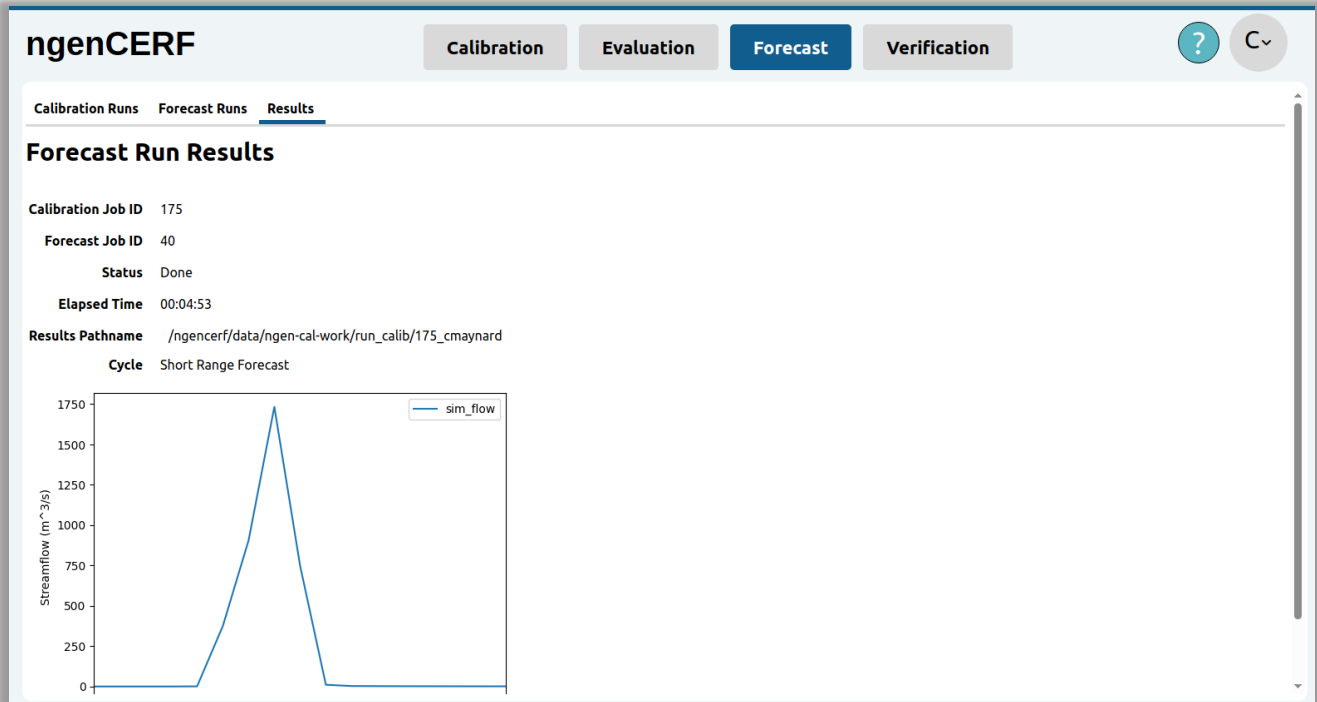
If status is Ready click Run to submit and run the forecast.

Forecast Job Run Time		Forecast Job Status	
Calibration Job ID	351	Status	Done
Forecast Job ID	49	Configuration	Short Range Forecast
Submit Time	Jun 23, 2025, 10:41:47		
Elapsed Time	00:01:28		
Results Pathname	/ngencerf/data/ngen-cal-work/run_calib/351_cmaynard		

View Results

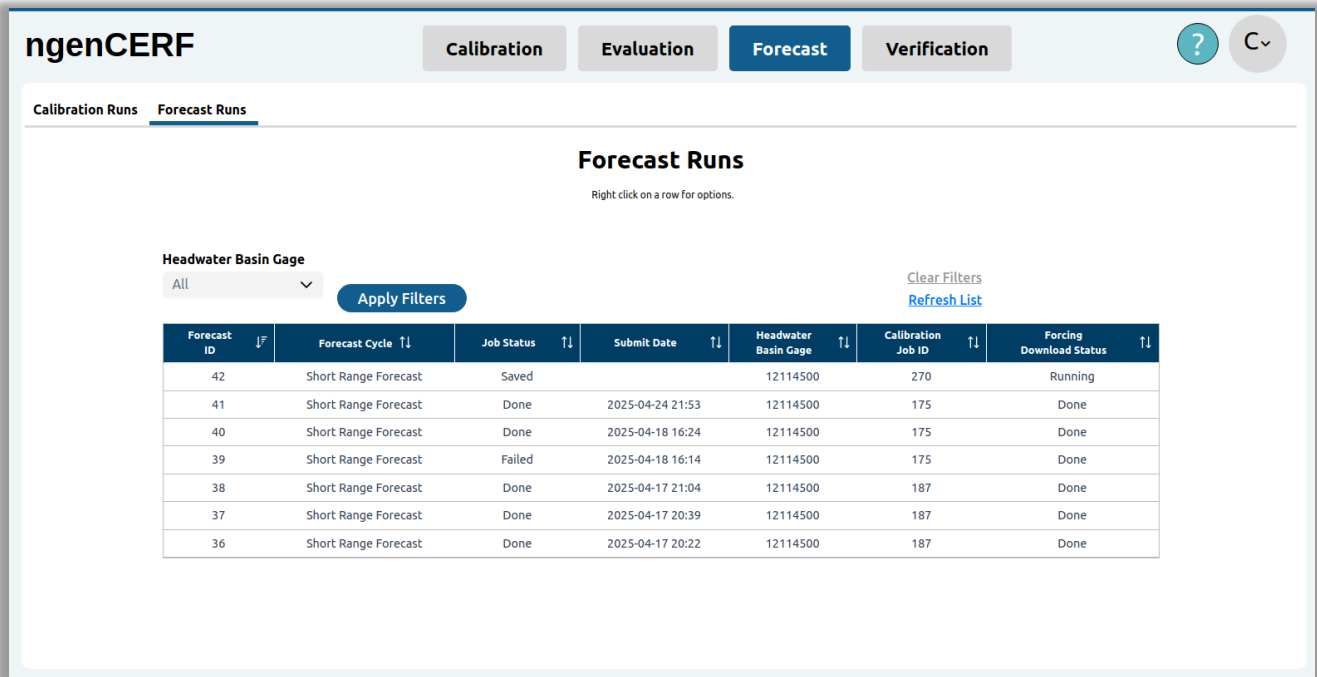
5.6.4 Results Tab

After the forecast is done, view any auto-generated output.



5.6.5 Forecast Runs Tab

List of forecast runs. This contains only 30 days of forecasts.



ngenCERF Calibration Evaluation **Forecast** Verification

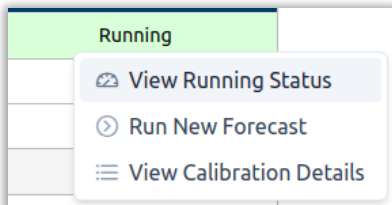
Calibration Runs **Forecast Runs**

Forecast Runs
Right click on a row for options.

Headwater Basin Gage: All [Clear Filters](#) [Refresh List](#) [Apply Filters](#)

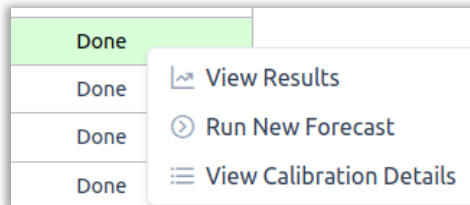
Forecast ID	Forecast Cycle	Job Status	Submit Date	Headwater Basin Gage	Calibration Job ID	Forcing Download Status
42	Short Range Forecast	Saved		12114500	270	Running
41	Short Range Forecast	Done	2025-04-24 21:53	12114500	175	Done
40	Short Range Forecast	Done	2025-04-18 16:24	12114500	175	Done
39	Short Range Forecast	Failed	2025-04-18 16:14	12114500	175	Done
38	Short Range Forecast	Done	2025-04-17 21:04	12114500	187	Done
37	Short Range Forecast	Done	2025-04-17 20:39	12114500	187	Done
36	Short Range Forecast	Done	2025-04-17 20:22	12114500	187	Done

Forecast Running Context Menu:



- Running
- View Running Status
- Run New Forecast
- View Calibration Details

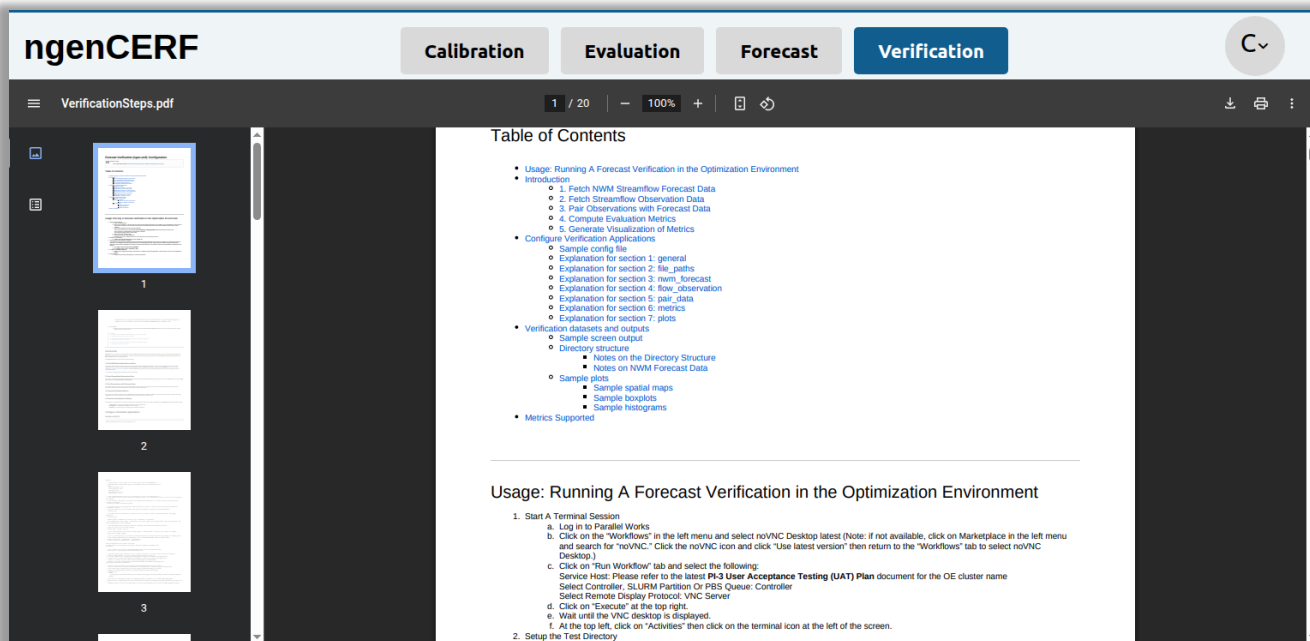
Forecast Done Context Menu:



- Done
- View Results
- Run New Forecast
- View Calibration Details

5.7 Verification Workflow

The Verification workflow consists of a PDF with detailed information and instructions for running the ngen-verf program.



6.0 External Data

The backend will retrieve data from Data Services, which retrieves it from external sources, for use within the GUI and the ngen app.

Data	Source Format	Ngen Format	Source
Forcing	netCDF or zarr	csv	NOAA AORC Dataset AWS S3 Bucket or Forcing Engine
Observational	Time series	csv	USGS National Water Information System
Subsetted Hydrofabric	gpkg	gpkg	NGWPC Hydrofabric Service API call

6.1 Forcing Data

The forcing data represents the forces affecting water input to a basin on an hourly basis (aka time series data). The following information is provided in the NOAA Analysis of Record for Calibration (AORC) dataset:

Table 6: Calibration Forcing Data Items

netCDF name	Zarr name*	Description*	Units
time	time	YYYY-MM-DD HH:00:00	
RAINRATE	APCP_surface	Hourly total precipitation	kg/m ²
T2D	TMP_2maboveground	Temperature at 2 m above ground level (AGL)	°K
Q2D	SPFH_2maboveground	Specific humidity at 2m AGL	kg/kg
PSFC	PRES_surface	Air pressure at surface	Pa
SWDOWN	DSWRF_surface	Downward shortwave (solar) radiation flux at the surface	W/m ²
LWDOWN	DLWRF_surface	Downward longwave (infrared) radiation flux at the surface	W/m ²
U2D	UGRD_10maboveground	U (west-east) component of the wind at 10 m AGL	m/s
V2D	VGRD_10maboveground	V (south-north) components of the wind at 10 m AGL	m/s

The ngen app expects the data to be in a comma delimited file for each headwater basement catchment associated with the gage.

Example: First few lines from aorc_nwm/cat-10617.csv

```
data > aorc_nwm > csv_basin_group1 > Gage_01123000 > cat-10617.csv > data
1  time,LWDOWN,PSFC,Q2D,RAINRATE,SWDOWN,T2D,U2D,V2D
2  2017-01-01 01:00:00,510.565436814698,99191.9671872943,0.0042,0.000112143136021398,0.277.3817030727,4.39607844384344,8.64363990515544
3  2017-01-01 02:00:00,304.057655192448,99123.7209400303,0.00426557767488709,0.000260330787711924,0.276.255007527034,4.50694542223108,8.41304797704547
4  2017-01-01 03:00:00,307.840746061239,98868.1567136485,0.00404472563367488,0.000180081722885704,0.275.901037957814,4.64419528157747,8.18862073657279
5  2017-01-01 04:00:00,301.80244049728,99031.8568623861,0.00400265617355005,0.000158074806470506,0.274.625961932181,4.98118912622364,8.94355216746247
```

6.2 Observational Data

The observational data represents the streamflow in a time series at a gage.

For the ngen app, it is expected to be in a comma delimited file (CSV) associated with the gage.

Example: First few lines from streamflow_obs/01123000_hourly_discharge.csv

```
data > streamflow_obs > 01123000_hourly_discharge.csv > data
1  dateTime,q_cms
2  2017-01-01 05:00:00,1.4894661522
3  2017-01-01 06:00:00,1.4894661522
4  2017-01-01 07:00:00,1.4894661522
5  2017-01-01 08:00:00,1.4894661522
```

6.3 Subsetted Hydrofabric

The backend will call Data Services to obtain the subsetted hydrofabric GeoPackage (gpkg) for the Calibration run. A GeoPackage is an open, standards-based, platform-independent, portable, self-describing, compact format for transferring geospatial information. A GeoPackage is the SQLite container, and it supports direct use. This means that data in a GeoPackage can be accessed and updated in a “native” storage format without intermediate format translations⁵.

To view the GeoPackage file, use a program like QGIS.

6.4 Example Calibratable Parameters Load File

This file can be a CSV or text file and must have a header line with param, min, max, init, and model. Each line in the file identifies the calibratable parameter name, its minimum value, its maximum value, its initial value and the model name. This information can be comma separated, space separated, or tab separated. The parameter and model names are case-sensitive.

Valid model names are:

CFE-S, CFE-X, LASAM, Noah-OWP-Modular, Sac-SMA, Snow-17, TopModel, UEB.

Example csv file:

```
param,min,max,init,model
field_capacity,10.5,500.0,20.0,LASAM
hydraulic_conductivity,1.0,50.0,10.0,LASAM
smcmax,0.4,0.7,0.439,LASAM
```

⁵ <https://www.ogc.org/standard/geopackage/>

6.5 Example Calibration Definition File

This JSON file can be exported using the ngenCERF CLI “export” command and used as a template to create a new calibration job using the ngenCERF CLI “import” command.

```
{
  "metadata": {
    "source_calibration_run_id": 269,
    "source_status": "Done",
    "time_range": {
      "start_time": "2013-01-01T00:00:00+00:00",
      "end_time": "2022-12-31T22:00:00+00:00"
    },
    "job_data_dir": "/ngencertif/data/ngen-cal-work/run_calib/269_peter",
    "num_catchments": 5
  },
  "run_after_import": false,
  "gage_id": "01055000",
  "forcing_source": "AORC",
  "observational_source": "Data Services",
  "geopackage_source": "User Upload",
  "geopackage_user_uploaded_file_path": "/ngencertif/data/ngen-cal-
work/run_calib/269_peter/geopackage/gauge_01123000.gpkg",
  "modules": [
    "T-Route",
    "CFE-S",
    "Noah-OWP-Modular"
  ],
  "formulation_name": "noah_cfes_troute_01055000_clone",
  "use_sloth": false,
  "sloth_parameters": [],
  "automatic_validation": true,
  "calibration_times": {
    "calibration_start_time": "2016-10-01T00:00:00Z",
    "calibration_end_time": "2017-09-01T00:00:00Z",
    "simulation_start_time": "2015-10-01T00:00:00Z",
    "simulation_end_time": "2017-09-01T00:00:00Z"
  },
  "validation_times": {
    "validation_start_time": "2015-10-01T00:00:00Z",
    "validation_end_time": "2016-09-01T00:00:00Z",
    "simulation_start_time": "2014-10-01T00:00:00Z",
    "simulation_end_time": "2017-09-01T00:00:00Z"
  },
  "streamflow_threshold": 3.88,
  "peak_flow_threshold": null,
  "parameters": [
    {
      "name": "b",
      "minimum": 2.0,
      "maximum": 15.0,
      "initial_value": 4.05,
      "module": "CFE-S"
    }
  ],
}
```

```
"objective_function": "KGE",  
"optimization_inputs": [  
  {  
    "name": "r",  
    "value": 0.2  
  }  
],  
"optimization": "DDS",  
"save_plot_iteration_frequency": 1,  
"save_output_iteration": false,  
"stop_criteria": 5  
}
```

7.0 Acronyms

Acronym	Definition
CFE-S	Conceptual Functional Equivalent Model - Schaake
CFE-X	Conceptual Functional Equivalent Model - Xinanjiang
CLI	Command Line Interface
CONUS	Continental United States
COR	Pearson Correlation
CSI	Critical Success Index
EDFS	Enterprise Data and FIM Services (Formally Enterprise Hydrofabric Services) aka Data Services in this document
EVBIAS	Event Volume Bias
EWTS	Error Warning and Trapping System
FAR	False Alarm Ratio
FBIAS	Frequency bias
FIM	Flood Inundation Mapping
GUI	Graphical User Interface
HFDC	Percent bias of high flow segment of flow duration curve
KGE	Kling-Gupta Efficiency
LASAM	Lumped Arid Semi-Arid Model
LFDC	Percent bias of low flow segment of flow duration curve
LogNSE	NSE of logarithmic values
LSTM	Long Short-Term Memory Networks Model
MAE	Mean Absolute Error
MFDC	Percent bias of middle flow segment of flow duration curve
ngenCERF	ngen Calibration, Evaluation, Reforecast, Forecast
NNSE	Normalized NSE
Noah	A metaphor for modeling water and land surface processes, particularly floods, rainfall, and soil water. The Noah model was developed by the NOAA/NCEP (National Centers for Environmental Prediction) starting in the 1990s.
Noah-MP	Noah-Multi-Physics
Noah-OWP-Modular	Extended, refactored version of the Noah-MP land surface model
NSE	Nash-Sutcliffe-Efficiency
NSEwt	Weighted NSE (NSE and LogNSE)
NWM	National Water Model
PBIAS	Percent Bias
PDF	Postscript Document Format
PET	Potential Evapotranspiration Model
PKBIAS	Absolute Peak Flow Bias
PKTE	Peak Flow Timing Error
POD	Probability of Detection
pPKBIAS	Percent peak flow bias
PW	Parallel Works
RMSE	Root Mean Square Error
RSR	Ratio of RMSE to standard deviation of observation
Sac-SMA	Sacramento Soil Moisture Accounting Model
SCHISM	Semi-implicit Cross-scale Hydrosience Integrated System Model
SFINCS	Super-fast modelling of coastal flooding events Model
SFT	Soil Freeze Thaw Model
SLoTH	Simple Logical Tautology Handler

SMP	Soil Moisture Profiles Model
Snow-17	Snow accumulation and melt Model
TBD	To Be Determined
TopModel	A physically based, distributed watershed Model
T-Route	Tree-Based Channel Routing Model
UEB	Utah Energy Balance Model